The Alaoglu–Erdős Conjecture

A Fourth Research Continuation

The adelic–HCIZ decomposition, exact progression determinants, integer-valued divisibility, and a finite exclusion beyond the inherited range

Research report prepared for Vladimir Reshetnikov

21 August 2026

Abstract

The Alaoglu–Erdős conjecture asks whether a real number x for which 2^x and 3^x are integers must itself be an integer. The conjecture is still open: it is a particularly concrete consequence of the Four Exponentials Conjecture, and the most recent directly relevant work on the Matrix Coefficient Conjecture leaves the decisive rank-two implication open. Accordingly, this report does *not* claim a completed proof.

It continues the attached unified report in five directions. First, it gives an exact finite decomposition of every exact-grid interpolation determinant into a geometric screen, an HCIZ entropy, and an adelic alignment defect. Second, it proves sharp no-go thresholds for the triangular determinant family: even an exact evaluation of the HCIZ integral and perfect joint prime transport cannot reach the already certified range. Third, it finds a new power-cusp family that crosses the universal geometric screen, and then proves why a separated-support valuation branch still defeats this family. Fourth, it factors a signed-lattice progression determinant completely into integer differences $P^d - Q^d$, exposing an unavoidable denominator/height tax in the continued-fraction strategy. Fifth, it proves a universal integer-valued factorial divisor and quantifies its subcritical $N^{3/2}\log N$ size. A directed-rounding MPFR computation independently extends the finite exclusion; the exact endpoint reached by the completed scan is stated in Section 12.

The report ends with three precise proof targets. Any one would settle the conjecture: an analytic fewnomial construction of the Dasgupta–Kakde type; a determinant family with a negative screen and controlled HCIZ/alignment terms; or a genuinely N^2-scale non-tropical valuation theorem. All computer-assisted statements are explicitly separated from paper proofs and from kernel-checked statements inherited from the supplied Lean corpus.

Outcome and trust boundary

No unconditional proof of the conjecture was obtained. Fabricating one would conceal, rather than solve, the Four Exponentials obstruction. The paper-level theorems in this report are fully proved here; numerical HCIZ values are labelled reconnaissance; and the finite exclusion is a software-level directed-rounding certificate, not a Lean kernel theorem.

Contents

1 Executive summary and claim ledger

1.1 The problem and the outcome

Conjecture 1.1 (Alaoglu–Erdős). *If $x \in \mathbb{R}$ and $2^x, 3^x \in \mathbb{Z}$, then $x \in \mathbb{Z}$.*

A negative exponent is impossible because $0 < 2^x < 1$. A rational exponent is also harmless: if $x = a/b$ in lowest terms and $2^x = M \in \mathbb{N}$, then $M^b = 2^a$, so $b \mid a$ and $b = 1$. If x were algebraic irrational, the Gelfond–Schneider theorem would make 2^x transcendental. Thus every hypothetical counterexample is positive, irrational, and transcendental.

The standard logarithmic matrix is

$$L = \begin{pmatrix} \log 2 & \log(2^x) \\ \log 3 & \log(3^x) \end{pmatrix} = \begin{pmatrix} \log 2 \\ \log 3 \end{pmatrix} (1 \quad x). \tag{1}$$

If x is irrational, the two factors have no nonzero rational annihilator. The Four Exponentials Conjecture predicts that a rank-one matrix of this form cannot have all four exponentials algebraic. This is exactly why the apparently elementary integer problem remains open; see Waldschmidt [9] and Dasgupta–Kakde [3].

1.2 What is new in this continuation

Principal paper-level results

1. The exact adelic–HCIZ sandwich

$$T_N \le C_N^- \le M_N \le \log I_N \le C_N^+$$

and the exact decomposition

$$\log(D_N/Q_N) = S_N + H_N + A_N.$$

Here Q_N is the termwise tropical divisor, S_N is a purely geometric screen, $H_N \ge 0$ is HCIZ entropy, and $A_N \ge 0$ is the loss caused by primewise assignments that cannot be aligned globally.

2. For the inherited triangular grids, the strongest conceivable joint tropical/HCIZ method is already impossible once

$$\beta > 6.9269442924\ldots.$$

The componentwise method is categorically impossible already beyond $\beta > 2.1520605288\ldots$. These are rigorous no-go thresholds, not merely failures of a particular numerical bound.

3. A family of power-cusp lattice regions has an explicit screen

$$S_q(z) = \tfrac{1}{2}\log(qz) + \tfrac{3}{4} - H_q - \tfrac{1}{2q} + qz\left[\left(\frac{q}{q+1}\right)^2 - B\left(1+\frac{1}{q}, 1+\frac{1}{q}\right)\right],$$

which tends to $-\infty$ for every fixed hypothetical exponent. Thus the triangular no-go is not universal. However, a separated-support valuation branch makes the componentwise deficit grow like $qz/9$, proving that cusp concentration alone cannot be a universal proof.

4. A signed-lattice determinant factors exactly as

$$|D_N| = (PQ)^{\binom{N}{3}} \prod_{d=1}^{N-1} |P^d - Q^d|^{N-d},$$

with

$$\log(P/Q) = (q\log 2 - p\log 3)(r - s\beta).$$

The factorization gives an exact "denominator tax": the small projected spacing supplied by two continued fractions is paid back by integer heights of order qs.

5. For every lower set of monomials, an integer evaluation determinant is divisible by the product of the corresponding factorials. On critical rank-two weighted triangles its logarithm is only $O(N^{3/2}\log N) = o(N^2)$, so ordinary integer-valued interpolation cannot supply the missing leading constant.

1.3 Claim ledger

Item	Content	Status
Conditional primitive generator	If a counterexample exists, the complete solution set is $\mathbb{N}_0 + \mathbb{N}_0\beta$, with integer output generators $(2,3)$ and (M,A)	Inherited; kernel-checked in the supplied corpus
Adelic–HCIZ decomposition	Finite exact identity and inequalities for every exact-grid determinant	Proved in this report
Triangular thresholds	$\beta_{\text{sep}} = 2.1520605\ldots$ and $\beta_{\text{joint}} = 6.9269443\ldots$	Proved in this report
Power-cusp screen	Explicit formula, asymptotic $S_q \to -\infty$, and separated-support obstruction	Proved in this report
HCIZ tables	High-precision finite-N evaluations for quantile discretizations	Numerical reconnaissance only
Signed progression determinant	Complete factorization and near-collision expansion	Proved in this report
Fewnomial support bound	A polynomial vanishing on a $2N \times 2N$ counterexample grid has at least $4N^2+1$ monomials	Proved in this report
Finite exclusion	Directed MPFR intervals on the new dyadic range; endpoint in Section 12	Software certificate, independently replayed
Full conjecture	$2^x, 3^x \in \mathbb{Z} \Rightarrow x \in \mathbb{Z}$	Open

2 Inherited exact counterexample model

The supplied report contains a long chain of arithmetic reductions and a substantial Lean corpus. Only the consequences needed below are restated. Write

$$\alpha = \log 2, \qquad \gamma = \log 3.$$

Theorem 2.1 (Inherited primitive-generator theorem)**.** *Assume the Alaoglu–Erdős conjecture is false. Then there is a unique least nonintegral solution $\beta > 0$. Put*

$$M = 2^{\beta} \in \mathbb{N}, \qquad A = 3^{\beta} \in \mathbb{N}.$$

The complete solution set and its two integer-output monoids are

$$\mathcal{S} = \{n + k\beta : n, k \in \mathbb{N}_0\},$$
$$\{2^x : x \in \mathcal{S}\} = \{2^n M^k : n, k \in \mathbb{N}_0\},$$
$$\{(2^x, 3^x) : x \in \mathcal{S}\} = \{(2^n M^k, 3^n A^k) : n, k \in \mathbb{N}_0\}.$$

All coordinate representations are unique.

The irrationality of β makes $n + k\beta$ distinct. Leastness also gives useful primitivity constraints: M and A are not simultaneous perfect powers of a common degree > 1, and one cannot have both $2 \mid M$ and $3 \mid A$. These conditions eliminate easy affine descents, but they do not control all external prime supports.

The exact integer grid behind the rest of the report is

$$(n, k) \longmapsto (2^n M^k, 3^n A^k), \qquad n, k \in \mathbb{N}_0. \tag{2}$$

Its logarithmic matrix is

$$L_{\beta} = \begin{pmatrix} \alpha & \beta\alpha \\ \gamma & \beta\gamma \end{pmatrix} = \begin{pmatrix} \alpha \\ \gamma \end{pmatrix} (1 \ \beta). \tag{3}$$

Both factors are $\mathbb{Q}$-linearly independent. Thus a counterexample is the smallest possible unresolved rank-two logarithm configuration: algebraicity is replaced by the stronger condition of integrality, but the determinant barrier is unchanged.

3 Exact-grid interpolation determinants

Let $\mathcal{R}, \mathcal{C} \subset \mathbb{N}_0^2$ have the same cardinality N. For $r = (a, b) \in \mathcal{R}$ and $c = (n, k) \in \mathcal{C}$, set

$$u_r = a\alpha + b\gamma, \qquad v_c = n + k\beta, \tag{4}$$

and

$$E_{r,c} = \exp(u_r v_c) = 2^{an} 3^{bn} M^{ak} A^{bk} \in \mathbb{N}. \tag{5}$$

After sorting the u- and v-values increasingly, write

$$D_N = \det(E_{ij})_{1 \le i,j \le N}.$$

Lemma 3.1 (Strict total positivity)**.** *For strictly increasing real vectors $u_1 < \cdots < u_N$ and $v_1 < \cdots < v_N$,*

$$\det(\mathrm{e}^{u_i v_j})_{i,j=1}^{N} > 0.$$

Consequently every determinant D_N in (5) is a positive integer.

Proof. The functions $f_j(t) = \mathrm{e}^{v_j t}$ form an extended complete Chebyshev system:

$$\det(f_j^{(i-1)}(t))_{i,j=1}^{N} = \mathrm{e}^{t \sum_j v_j} \prod_{i<j} (v_j - v_i) > 0.$$

The generalized mean-value theorem for Chebyshev systems transfers the sign of this Wronskian to every evaluation determinant at increasing nodes. The entries in (5) are integers, so the determinant is a positive integer. □

Let

$$\Delta(u) = \prod_{i<j}(u_j - u_i), \qquad G(N+1) = \prod_{j=1}^{N-1} j!.$$

The Harish–Chandra–Itzykson–Zuber identity gives

$$I_N(u,v) := \int_{U(N)} \exp\bigl(\mathrm{Tr}(\mathrm{diag}(u)U\,\mathrm{diag}(v)U^*)\bigr)\,\mathrm{d}U = G(N+1)\frac{D_N}{\Delta(u)\,\Delta(v)}, \tag{6}$$

where Haar measure is normalized to mass one. Equivalently,

$$\log D_N = V_N + \log I_N, \qquad V_N := \log\Delta(u) + \log\Delta(v) - \log G(N+1). \tag{7}$$

For each prime p, define the entry cost

$$w_{p,ij} = v_p(E_{ij}) = \mathbf{1}_{p=2}a_in_j + \mathbf{1}_{p=3}b_in_j + v_p(M)a_ik_j + v_p(A)b_ik_j, \tag{8}$$

and its tropical assignment minimum

$$\tau_p = \min_{\sigma\in S_N}\sum_{i=1}^{N} w_{p,i\sigma(i)}. \tag{9}$$

The nonarchimedean triangle inequality gives

$$v_p(D_N) \geq \tau_p. \tag{10}$$

Only finitely many primes occur in the entries.

4 The exact adelic–HCIZ decomposition

Define the reverse, independent, and forward real assignment costs

$$C_N^- := \sum_{i=1}^{N} u_iv_{N+1-i}, \tag{11}$$

$$M_N := \frac{(\sum_i u_i)(\sum_j v_j)}{N}, \tag{12}$$

$$C_N^+ := \sum_{i=1}^{N} u_iv_i. \tag{13}$$

Also put

$$T_N := \sum_p \tau_p \log p, \qquad Q_N := \prod_p p^{\tau_p}. \tag{14}$$

Because all entries are integers, Q_N is a positive integer dividing every Leibniz term and hence dividing D_N.

Theorem 4.1 (Adelic–HCIZ sandwich)**.** *For every exact-grid determinant,*

$$\boxed{T_N \leq C_N^- \leq M_N \leq \log I_N \leq C_N^+.} \tag{15}$$

Proof. For each entry, unique factorization gives

$$u_iv_j = \log E_{ij} = \sum_p w_{p,ij}\log p.$$

Therefore the sum of the primewise minima is no larger than the minimum of the sum:

$$\begin{aligned} T_N &= \sum_p (\log p) \min_\sigma \sum_i w_{p,i\sigma(i)} \\ &\leq \min_\sigma \sum_i \sum_p w_{p,i\sigma(i)} \log p = \min_\sigma \sum_i u_i v_{\sigma(i)} = C_N^-. \end{aligned}$$

The final equality is the rearrangement inequality. Averaging the assignment cost over all permutations gives M_N, so $C_N^- \leq M_N \leq C_N^+$.

For the HCIZ integral, if $Z(U) = \mathrm{Tr}(\mathrm{diag}(u) U \,\mathrm{diag}(v) U^*)$, then

$$\mathbb{E}Z = \frac{\mathrm{Tr}(\mathrm{diag}\, u)\,\mathrm{Tr}(\mathrm{diag}\, v)}{N} = M_N.$$

Jensen's inequality gives $\log I_N \geq M_N$. Von Neumann's trace inequality gives $Z(U) \leq C_N^+$ for every U, hence $\log I_N \leq C_N^+$. □

Definition 4.2 (Three exact defects)**.** Set

$$\Xi_N := \log(D_N/Q_N), \tag{16}$$
$$S_N := V_N + M_N - C_N^-, \tag{17}$$
$$H_N := \log I_N - M_N, \tag{18}$$
$$A_N := C_N^- - T_N. \tag{19}$$

We call these, respectively, the cancellation/new-prime excess, the geometric screen, the HCIZ entropy, and the adelic alignment defect.

Theorem 4.3 (Exact three-term decomposition)**.** *For every exact-grid determinant,*

$$\boxed{\Xi_N = S_N + H_N + A_N.} \tag{20}$$

Moreover

$$\Xi_N \geq 0, \qquad H_N \geq 0, \qquad A_N \geq 0.$$

In fact D_N/Q_N is a positive integer, so either $\Xi_N = 0$ or $\Xi_N \geq \log 2$.

Proof. Equation (7) and the definitions give

$$\log D_N - T_N = (V_N + M_N - C_N^-) + (\log I_N - M_N) + (C_N^- - T_N).$$

The left side is $\log(D_N/Q_N)$. Nonnegativity of H_N and A_N is Theorem 4.1; nonnegativity and discreteness of Ξ_N follow from $Q_N \mid D_N$. □

The decomposition separates three logically different tasks. Geometry can make S_N negative. Random-matrix entropy pushes upward by H_N. Primewise assignments that prefer incompatible permutations push upward by A_N. A determinant proof based only on termwise valuations must produce a family for which the negative screen dominates the other two terms.

Corollary 4.4 (Equality criterion for adelic alignment)**.** *Assume u and v are strictly increasing. Then $A_N = 0$ if and only if the reverse permutation $i \mapsto N+1-i$ minimizes every prime cost occurring in the matrix.*

Proof. For every prime, the reverse cost is at least τ_p. Their logarithmically weighted sum is C_N^-. Equality $T_N = C_N^-$ holds exactly when every nonnegative difference vanishes. Strict rearrangement makes the reverse permutation the unique minimizer of the summed real cost. □

4.1 A finite permanent refinement

The HCIZ upper bound has a useful exact refinement, although it does not improve the N^2 asymptotic scale.

Proposition 4.5 (Birkhoff–permanent upper bound)**.** *For arbitrary real vectors u, v,*

$$I_N(u,v) \le \frac{1}{N!}\,\mathrm{perm}(\mathrm{e}^{u_i v_j})_{i,j=1}^{N}. \tag{21}$$

Consequently

$$\log I_N \le C_N^+, \qquad 0 \le C_N^+ - \log\left(\frac{\mathrm{perm}(\mathrm{e}^{u_i v_j})}{N!}\right) \le \log N!.$$

Thus the permanent bound and the von Neumann bound have the same normalized N^2 limit.

Proof. For Haar U, the matrix $P(U) = (|U_{ij}|^2)$ is doubly stochastic. For each doubly stochastic P, choose the unique minimum-Euclidean-norm Birkhoff decomposition $P = \sum_\sigma \lambda_\sigma(P) P_\sigma$. Uniqueness makes this choice equivariant under independent row and column permutations. Convexity gives

$$\exp\langle u, Pv\rangle \le \sum_\sigma \lambda_\sigma(P) \exp\left(\sum_i u_i v_{\sigma(i)}\right).$$

The law of $P(U)$ is invariant under the permutation action, which is transitive on S_N. Hence $\mathbb{E}\lambda_\sigma(P(U)) = 1/N!$ for every σ. Integration proves (21). The largest permanent term is $\mathrm{e}^{C_N^+}$; using it once and bounding all terms by it yields the final inequalities. □

4.2 Rational gauge invariance

The same local quotient survives rational changes of basis, where individual entries may acquire denominators.

Proposition 4.6 (Rational local quotient)**.** *Let $E = (E_{ij})$ be a positive rational matrix with nonzero determinant D. For each prime set*

$$\tau_p = \min_\sigma \sum_i v_p(E_{i\sigma(i)}), \qquad Q = \prod_p p^{\tau_p}.$$

Then $D/Q \in \mathbb{Z} \setminus \{0\}$. Multiplying arbitrary rows and columns by positive rational numbers changes D and Q by the same factor, so D/Q is invariant.

Proof. Every Leibniz term has p-adic valuation at least τ_p, hence $v_p(D/Q) \ge 0$ for all p. A nonzero rational number with nonnegative valuation at every prime is an integer. A row multiplier adds the same local potential to every assignment and to the determinant valuation; column multipliers behave identically. □

This proposition is the correct bookkeeping rule for signed or continued-fraction bases: denominators are not an external nuisance that may be discarded. They are local gauge factors, and the invariant quotient records the exact amount that remains after they are paid.

5 The triangular family: sharp ceilings for the inherited method

The inherited determinant construction uses the first lattice points in the positive quadrant, ordered by the two linear forms in (4). This section computes every leading constant exactly and separates three levels of information:

(i) the old componentwise tropical divisor together with the crude comonotone bound $\log I_N \le C_N^+$;

(ii) the same componentwise divisor together with a hypothetical exact evaluation of the HCIZ free energy;

(iii) the best termwise tropical divisor permitted by the joint prime costs, again together with a hypothetical exact HCIZ evaluation.

The third level is an absolute ceiling for every proof that uses this family and only termwise determinant valuations.

5.1 Limiting spectra and logarithmic energy

Let $\mathcal{R}_N$ consist of the first N points $(a,b) \in \mathbb{N}_0^2$ in increasing order of $a\alpha + b\gamma$, and let $\mathcal{C}_N$ consist of the first N points (n,k) in increasing order of $n + k\beta$. Boundary ties do not occur because α/γ and β are irrational. Set

$$U = \sqrt{2\alpha\gamma}, \qquad V = \sqrt{2\beta}, \qquad c = UV = 2\sqrt{\beta\alpha\gamma}. \tag{22}$$

Standard lattice-point counting in a triangle gives, uniformly away from the endpoints,

$$\frac{u_i}{\sqrt{N}} \longrightarrow U\sqrt{s}, \qquad \frac{v_i}{\sqrt{N}} \longrightarrow V\sqrt{s}, \qquad s = \frac{i}{N}. \tag{23}$$

Equivalently, the limiting measures have densities $2t/U^2$ on $[0,U]$ and $2t/V^2$ on $[0,V]$.

Lemma 5.1 (Energy of the triangular radial law). *If X, Y are independent with density $2t$ on $[0,1]$, then*

$$\mathbb{E}\log|X-Y| = -\frac{7}{4}. \tag{24}$$

Consequently the geometric Vandermonde term satisfies

$$\frac{V_N}{N^2} = \frac{1}{2}\log c - 1 + o(1). \tag{25}$$

Proof. By symmetry and the substitution $y = xt$,

$$\begin{aligned}
\mathbb{E}\log|X-Y| &= 8\int_0^1 x\int_0^x y\log(x-y)\,\mathrm{d}y\,\mathrm{d}x \\
&= 8\int_0^1 x^3\int_0^1 t(\log x + \log(1-t))\,\mathrm{d}t\,\mathrm{d}x = -\frac{7}{4}.
\end{aligned}$$

For a compact atomless limiting measure μ, the usual truncated-kernel argument gives

$$\frac{1}{N^2}\log\Delta(\sqrt{N}x_1,\dots,\sqrt{N}x_N) = \frac{1}{4}\log N + \frac{1}{2}\iint \log|x-y|\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) + o(1).$$

The Barnes asymptotic

$$\log G(N+1) = \frac{1}{2}N^2\log N - \frac{3}{4}N^2 + o(N^2)$$

cancels the two $\frac{1}{4}N^2 \log N$ terms. Scaling (24) by U and V gives

$$\frac{1}{2}(\log U - \tfrac{7}{4} + \log V - \tfrac{7}{4}) + \frac{3}{4} = \frac{1}{2}\log c - 1.$$

□

The three real assignment costs are equally explicit. Since the quantile is $\sqrt{s}$,

$$\frac{M_N}{N^2} = \frac{4c}{9} + o(1), \tag{26}$$

$$\frac{C_N^-}{N^2} = c\int_0^1 \sqrt{s(1-s)}\,\mathrm{d}s + o(1) = \frac{\pi c}{8} + o(1), \tag{27}$$

$$\frac{C_N^+}{N^2} = c\int_0^1 s\,\mathrm{d}s + o(1) = \frac{c}{2} + o(1). \tag{28}$$

5.2 The componentwise tropical divisor

Write the standard simplex coordinates

$$X = \frac{\alpha a}{U\sqrt{N}}, \quad Y = \frac{\gamma b}{U\sqrt{N}}, \qquad P = \frac{n}{V\sqrt{N}}, \quad Q = \frac{\beta k}{V\sqrt{N}}.$$

Each pair is asymptotically uniform on $x + y \le 1$. Every one-coordinate marginal has density $2(1-t)$ and quantile

$$q_\triangle(s) = 1 - \sqrt{1-s}.$$

Its countermonotone product is

$$J := \int_0^1 q_\triangle(s) q_\triangle(1-s)\,\mathrm{d}s = \frac{\pi}{8} - \frac{1}{3}. \tag{29}$$

The four logarithmic rank-one components

$$\alpha a n, \qquad \gamma b n, \qquad \beta\alpha a k, \qquad \beta\gamma b k$$

each contribute $cJN^2 + o(N^2)$ to the sum of their separate assignment minima. Since the minimum of a sum is at least the sum of the separate minima, the universal componentwise bound is

$$T_N \ge T_N^{\mathrm{sep}}, \qquad \frac{T_N^{\mathrm{sep}}}{N^2} = 4cJ + o(1). \tag{30}$$

Equality holds when the four prime supports are pairwise disjoint; this important branch returns in Section 7.4.

5.3 Three exact no-go thresholds

Define

$$\delta_{\mathrm{old}} = \frac{11}{6} - \frac{\pi}{2}, \qquad f_{\mathrm{old}}(c) = \frac{1}{2}\log c - 1 + \delta_{\mathrm{old}} c, \tag{31}$$

$$\delta_{\mathrm{sep}} = \frac{16}{9} - \frac{\pi}{2}, \qquad f_{\mathrm{sep}}(c) = \frac{1}{2}\log c - 1 + \delta_{\mathrm{sep}} c, \tag{32}$$

$$\delta_{\mathrm{joint}} = \frac{4}{9} - \frac{\pi}{8}, \qquad f_{\mathrm{joint}}(c) = \frac{1}{2}\log c - 1 + \delta_{\mathrm{joint}} c. \tag{33}$$

The first is the normalized upper bound $(V_N + C_N^+ - T_N^{\mathrm{sep}})/N^2$. The second is the normalized lower floor $(V_N + M_N - T_N^{\mathrm{sep}})/N^2$: even perfect knowledge of the HCIZ integral cannot go below it. The third is the geometric screen $(V_N + M_N - C_N^-)/N^2$, corresponding simultaneously to the smallest possible HCIZ value and the largest possible termwise tropical divisor.

Theorem 5.2 (Sharp triangular ceilings). *For $\delta > 0$, the unique positive root of $\frac{1}{2}\log c - 1 + \delta c = 0$ is*

$$c_\delta = \frac{W(2\delta e^2)}{2\delta}, \qquad \beta_\delta = \frac{c_\delta^2}{4\alpha\gamma}, \tag{34}$$

where W is the principal Lambert function. For the three constants above,

method	δ	c_δ	β_δ
componentwise plus C_N^+	0.262537006538437	2.257900214994267	1.673707587366710
componentwise plus exact HCIZ floor	0.206981450982881	2.560307883957848	2.152060528880198
perfect joint tropical plus exact HCIZ floor	0.0517453627457203	4.593416223613747	6.926944292408383

In particular, for every $\beta > 6.926944292408383\ldots$, the triangular family has $S_N > 0$ for all sufficiently large N. Since $\Xi_N = S_N + H_N + A_N$ with $H_N, A_N \geq 0$, no asymptotic contradiction can be obtained from this family by any combination of an exact HCIZ evaluation and termwise prime valuations.

Proof. The derivative of $\frac{1}{2}\log c - 1 + \delta c$ is positive, so the root is unique. Exponentiating $\log c - 2 + 2\delta c = 0$ gives $(2\delta c)e^{2\delta c} = 2\delta e^2$, proving (34). Equations (25)–(30) produce the three displayed functions. The final assertion is exactly the nonnegative decomposition (20). □

At the inherited kernel-checked floor $\beta \geq 12$, and even more strongly at the software-certified range of Section 12, the best joint screen is already far positive:

β	c	$f_{\text{sep}}(c)$	$f_{\text{joint}}(c)$
12	6.0458250471	1.1510576201	0.2125273896
27	9.0687375706	1.9794769945	0.5716816488
28	9.2351503056	2.0230132549	0.5893846472
29	9.3986169838	2.0656206551	0.6066161197

The maximum normalized improvement obtainable by replacing the forward bound C_N^+ with the Jensen floor M_N is only

$$\frac{C_N^+ - M_N}{N^2} \longrightarrow \frac{c}{18}. \tag{35}$$

Thus "compute the HCIZ limit exactly" is not, by itself, a viable completion of the inherited triangular argument.

6 HCIZ free-energy reconnaissance

The preceding no-go theorem does not require the exact HCIZ free energy. Nevertheless, numerical evaluation is useful for measuring how much entropy is actually present and for calibrating new geometries.

6.1 A stable determinant formula

For triangular quantile nodes

$$x_i = \sqrt{\frac{i + 1/2}{N}}, \qquad 0 \leq i < N,$$

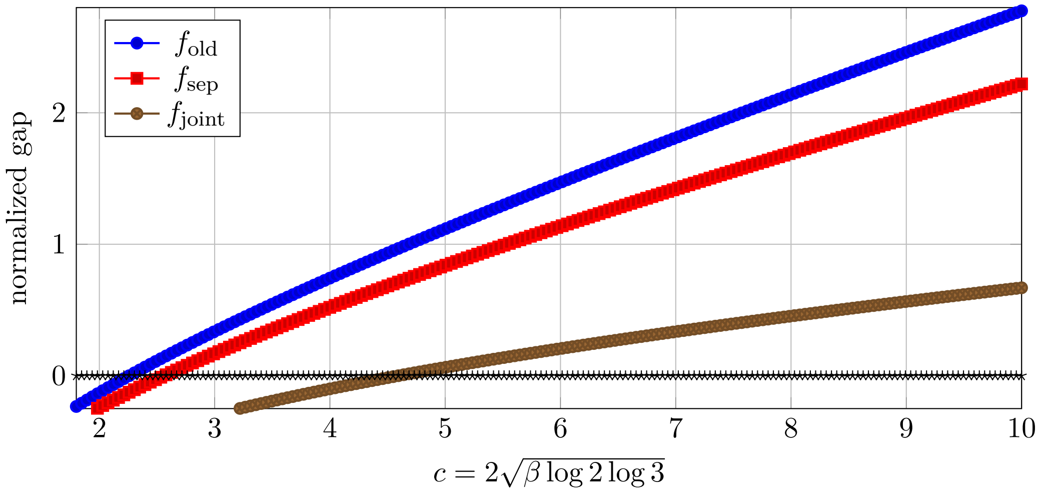

Figure 1: The three triangular gap functions. The joint screen crosses zero at $c = 4.5934162\ldots$, corresponding to $\beta = 6.9269443\ldots$.

and scaled spectra $u_i = \sqrt{N}\, Ux_i$, $v_i = \sqrt{N}\, Vx_i$, put $t = Nc$. The HCIZ identity becomes

$$I_N = G(N+1)t^{-N(N-1)/2}\frac{\det(e^{tx_ix_j})_{0\le i,j<N}}{\Delta(x)^2}. \tag{36}$$

High precision is essential: the numerator determinant is totally positive but extremely ill-conditioned in ordinary floating-point arithmetic.

c	$N=8$	$N=12$	$N=16$	$N=20$	Jensen $4c/9$	forward $c/2$
2.1972245773	0.9902881602	0.9875417092	0.9863697075	0.9857446579	0.9765442566	1.0986122887
2.5603078840	1.1552417262	1.1521074196	1.1507724885	1.1500614614	1.1379146151	1.2801539420
4.5934162236	2.0855796790	2.0806140315	2.0785278197	2.0774272195	2.0415183216	2.2967081118
9.0687375706	4.1684482612	4.1609010095	4.1578193572	4.1562264719	4.0305500314	4.5343687853

All entries in the four middle columns are $N^{-2}\log I_N$ computed from (36). They lie strictly between the Jensen and forward bounds, as they must. The table is numerical reconnaissance, not a proof of convergence or a certified enclosure.

6.2 An exact variance identity

Lemma 6.1 (Haar trace variance)**.** *Let A, B be Hermitian $N \times N$ matrices and put $A_0 = A - (\operatorname{Tr} A/N)I$, $B_0 = B - (\operatorname{Tr} B/N)I$. If U is Haar distributed on $U(N)$, then*

$$\operatorname{Var}(\operatorname{Tr}(AUBU^*)) = \frac{\operatorname{Tr}(A_0^2)\operatorname{Tr}(B_0^2)}{N^2-1}. \tag{37}$$

Proof. Unitary invariance reduces the covariance tensor of UB_0U^* to a linear combination of the identity and flip tensors. The conditions $\operatorname{Tr} B_0 = 0$ and $\mathbb{E}\operatorname{Tr}((UB_0U^*)^2) = \operatorname{Tr}(B_0^2)$ determine the coefficients; contraction with $A_0 \otimes A_0$ gives (37). Equivalently, it follows immediately from the second-order unitary Weingarten formula

$$\mathbb{E}(U_{ij}\overline{U_{kl}}U_{mn}\overline{U_{rs}}) = \frac{\delta_{ik}\delta_{mr}\delta_{jl}\delta_{ns} + \delta_{ir}\delta_{mk}\delta_{js}\delta_{nl}}{N^2-1} - \frac{\delta_{ik}\delta_{mr}\delta_{js}\delta_{nl} + \delta_{ir}\delta_{mk}\delta_{jl}\delta_{ns}}{N(N^2-1)}.$$

□

For the triangular law, $\mathrm{Var}(X) = 1/18$. Therefore the second Taylor coefficient of the centered HCIZ logarithm satisfies

$$\frac{1}{2N^2}\,\mathrm{Var}(\mathrm{Tr}(AUBU^*)) \longrightarrow \frac{c^2}{648}. \tag{38}$$

More explicitly, after writing the amplitude as a scalar parameter λ, the finite-N identity is

$$\frac{1}{N^2}\log\mathbb{E}\exp(\lambda Z_N) = \lambda\frac{M_N}{N^2} + \frac{\lambda^2}{2N^2}\,\mathrm{Var}(Z_N) + O_N(\lambda^3).$$

Thus the Jensen floor is not asymptotically exact even infinitesimally: the first entropy correction is positive and of order N^2.

Interpretation

The exact variance calculation explains the numerical gap but does not create a proof route. In the triangular family the geometric screen is already positive for every hypothetical exponent left by the inherited results. Any positive entropy only worsens that obstruction.

7 A general screen functional and power-cusp geometries

The triangular obstruction is a statement about one limiting spectral shape, not about all exact-grid determinants. This section derives the general screen functional and then constructs a family for which that functional is negative. The construction is a genuine advance: it proves that the positive triangular screen can be escaped. It is not yet a proof, because the adelic alignment term can be large.

7.1 The continuum screen

For a compactly supported atomless probability measure μ on $[0,\infty)$, write Q_μ for its increasing quantile, m_μ for its mean, and

$$\mathcal{E}(\mu) = \iint \log|x-y|\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \tag{39}$$

assuming the logarithmic energy is finite.

Theorem 7.1 (Continuum geometric screen)**.** *Suppose exact-grid spectra satisfy*

$$\frac{u_i}{\sqrt{N}} \Longrightarrow \mu, \qquad \frac{v_i}{\sqrt{N}} \Longrightarrow \nu,$$

with convergence strong enough for the logarithmic energies and first moments to converge. Then

$$\frac{S_N}{N^2} \longrightarrow \mathscr{S}(\mu,\nu) := \frac{1}{2}\mathcal{E}(\mu) + \frac{1}{2}\mathcal{E}(\nu) + \frac{3}{4} + m_\mu m_\nu - \int_0^1 Q_\mu(s)Q_\nu(1-s)\,\mathrm{d}s. \tag{40}$$

Proof. The two Vandermonde limits and the Barnes asymptotic are exactly those used in Lemma 5.1. The product of empirical means converges to $m_\mu m_\nu$. Finally, the rearrangement theorem identifies the reverse assignment limit with the countermonotone transport integral. Substitution in $S_N = V_N + M_N - C_N^-$ proves (40). □

The theorem turns the geometric part of the determinant problem into a variational question. A necessary condition for a termwise-valuation proof is a pair of realizable spectral measures with $\mathscr{S} < 0$. This condition is not sufficient: one must still control $H_N + A_N$.

7.2 A family of realizable cusps

Fix an integer $q \geq 2$. In the first-quadrant log-coordinate plane write

$$x = t\lambda, \qquad y = t(1-\lambda),$$

so that the Jacobian is t. Define

$$\mathcal{R}_q = \left\{ 0 \leq t \leq 1 : \left| \lambda - \frac{1}{2} \right| \leq \frac{1}{2} t^{q-2} \right\}. \tag{41}$$

The region is a narrowing cusp around the diagonal $x = y$, and

$$|\mathcal{R}_q| = \int_0^1 t\, t^{q-2}\, \mathrm{d}t = \frac{1}{q}. \tag{42}$$

For the row lattice, use the physical coordinates $x = \alpha a/(U_q\sqrt{N})$, $y = \gamma b/(U_q\sqrt{N})$; for the column lattice use $x = n/(V_q\sqrt{N})$, $y = \beta k/(V_q\sqrt{N})$, where

$$U_q = \sqrt{q\alpha\gamma}, \qquad V_q = \sqrt{q\beta}, \qquad c_q = U_q V_q = qz, \qquad z = \sqrt{\beta\alpha\gamma}. \tag{43}$$

Lattice Riemann sums in dilates of (41) produce exact lattice subsets up to $o(N)$ boundary modifications; such modifications do not affect any N^2 limit below. The radial coordinate has density

$$\rho_q(t) = q t^{q-1}, \qquad 0 \leq t \leq 1, \tag{44}$$

because the cross-sectional area element is $t^{q-1}\,\mathrm{d}t$.

Lemma 7.2 (Exact cusp energy and transport)**.** *Let T_q have density (44). Then*

$$m_q := \mathbb{E} T_q = \frac{q}{q+1}, \tag{45}$$

$$\mathcal{E}_q := \mathbb{E} \log |T_q - T_q'| = -H_q - \frac{1}{2q}, \tag{46}$$

$$B_q := \int_0^1 Q_q(s) Q_q(1-s)\, \mathrm{d}s = \mathrm{B}\left(1 + \frac{1}{q}, 1 + \frac{1}{q}\right), \tag{47}$$

where $H_q = \sum_{j=1}^{q} j^{-1}$ and $Q_q(s) = s^{1/q}$.

Proof. The mean and quantile are immediate. For the energy, symmetry and the substitution $y = xt$ give

$$\begin{aligned}
\mathcal{E}_q &= 2q^2 \int_0^1 x^{q-1} \int_0^x y^{q-1} \log(x-y)\, \mathrm{d}y\, \mathrm{d}x \\
&= 2q^2 \int_0^1 x^{2q-1}\, \mathrm{d}x \int_0^1 t^{q-1} (\log x + \log(1-t))\, \mathrm{d}t \\
&= -\frac{1}{2q} + q \int_0^1 t^{q-1} \log(1-t)\, \mathrm{d}t = -\frac{1}{2q} - H_q.
\end{aligned}$$

The last formula follows from the beta integral. ∎

Theorem 7.3 (Power-cusp screen)**.** *For the cusp families above,*

$$\boxed{\mathscr{S}_q(z) = \frac{1}{2}\log(qz) + \frac{3}{4} - H_q - \frac{1}{2q} + qz\left[\left(\frac{q}{q+1}\right)^2 - \mathrm{B}\left(1 + \frac{1}{q}, 1 + \frac{1}{q}\right)\right].} \tag{48}$$

For every fixed $z > 0$,

$$\mathscr{S}_q(z) = -\frac{1}{2}\log q + \frac{1}{2}\log z + \frac{3}{4} - \gamma_E + o(1) \longrightarrow -\infty. \tag{49}$$

Thus realizable exact-grid geometries with a negative joint screen exist for every hypothetical counterexample.

Proof. Scale the law of T_q by U_q and V_q in Theorem 7.1. The two energy terms contribute $\frac{1}{2}\log(qz) + \mathcal{E}_q$, the mean term is qzm_q^2, and the reverse transport is qzB_q. This proves (48).

Put $\varepsilon = 1/q$. The standard gamma expansion gives

$$\left(\frac{q}{q+1}\right)^2 = 1 - 2\varepsilon + 3\varepsilon^2 + O(\varepsilon^3),$$
$$\mathrm{B}(1+\varepsilon, 1+\varepsilon) = 1 - 2\varepsilon + \left(4 - \tfrac{\pi^2}{6}\right)\varepsilon^2 + O(\varepsilon^3).$$

Hence their difference is $(\pi^2/6 - 1)q^{-2} + O(q^{-3})$. Together with $H_q = \log q + \gamma_E + 1/(2q) + O(q^{-2})$, this yields (49). $\square$

For the two finite endpoints relevant to the computation, the screen crosses zero already at $q = 9$:

q	$m_q^2 - B_q$	$\mathscr{S}_q(z_{27})$	$\mathscr{S}_q(z_{28})$
2	0.0517453627	0.5716816488	0.5893846472
4	0.0219751076	0.3892297577	0.4056355442
6	0.0118358913	0.1903991210	0.2053999611
8	0.0073547992	0.0320015476	0.0459891876
9	0.0060106952	−0.0347761982	−0.0211831341
10	0.0050029689	−0.0949797019	−0.0817250021
12	0.0036203266	−0.1995903265	−0.1868836047
16	0.0021453252	−0.3641985599	−0.3522505734

Here $z_b = \sqrt{b \log 2 \log 3}$. The negative entries show exactly how much room is available for $H_N + A_N$; they do not assert that either defect is small.

7.3 HCIZ entropy in the cusp family

For quantile nodes $x_i = ((i + 1/2)/N)^{1/q}$, formula (36) remains valid with c replaced by $c_q = qz$. At $\beta = 28$ and $N = 32$, high-precision reconnaissance gives:

q	H_N/N^2	variance prediction $\frac{1}{2}c_q^2\,\mathrm{Var}(T_q)^2$	screen $\mathscr{S}_q(z_{28})$
9	0.0662792945	0.0578072909	−0.0211831341
10	0.0584731296	0.0505667121	−0.0817250021
12	0.0460717253	0.0394905746	−0.1868836047
16	0.0299562636	0.0258188672	−0.3522505734
24	0.0147022573	0.0133947985	−0.5814338175

The variance entry is exact at second order because

$$\mathrm{Var}(T_q) = \frac{q}{(q+1)^2(q+2)}, \qquad \frac{1}{2}c_q^2\,\mathrm{Var}(T_q)^2 = \frac{q^4 z^2}{2(q+1)^4(q+2)^2}. \tag{50}$$

The entropy decreases with q in this experiment, much more slowly than the screen at first. For $q \geq 10$ the numerical screen has enough room to absorb the observed HCIZ entropy. The remaining and more serious issue is the adelic alignment defect.

7.4 The separated-support obstruction

The inherited arithmetic constraints do not exclude

$$\gcd(M,6)=\gcd(A,6)=\gcd(M,A)=1. \tag{51}$$

Under (51), the supports of the four logarithmic rank-one components are pairwise disjoint. Primewise minimization therefore separates exactly.

Theorem 7.4 (Cusp failure in the separated-support branch)**.** *Assume (51). For the power-cusp grids,*

$$\frac{T_N}{N^2}=4c_qK_q+o(1), \tag{52}$$

where K_q is the countermonotone product of either coordinate marginal in $\mathcal{R}_q$. Moreover

$$K_q\longrightarrow\frac{2}{9},\qquad \frac{M_N-T_N}{N^2}=qz\left(m_q^2-4K_q\right)+o(1)\sim\frac{qz}{9}. \tag{53}$$

Consequently

$$\frac{V_N+M_N-T_N}{N^2}\longrightarrow+\infty\qquad(q\to\infty).$$

Thus power-cusp concentration cannot yield a universal proof by separate termwise valuations.

Proof. Under (51), primes dividing 2, 3, M, and A see respectively the costs an, bn, ak, and bk, up to their fixed logarithmic weights. Each cost has its own one-dimensional reverse matching, so equality holds in the componentwise separation. Symmetry of $\mathcal{R}_q$ makes all four normalized minima equal to c_qK_q.

Conditioned on the radial variable $T=t$, either coordinate has the exact representation

$$X_q=\frac{t}{2}+\frac{t^{q-1}}{2}Z,\qquad Z\sim\mathrm{Unif}[-1,1]. \tag{54}$$

As $q\to\infty$, $T\to1$ in probability and T^{q-1} converges in law to an independent $S\sim\mathrm{Unif}[0,1]$. Hence

$$X_q\Longrightarrow X_\infty=\frac{1+SZ}{2}.$$

The limit law is symmetric about $1/2$, has mean $1/2$, and has variance $\mathbb{E}(S^2)\mathbb{E}(Z^2)/4=1/36$. For a distribution symmetric about $1/2$, the countermonotone partner of X is $1-X$, so

$$K_q\longrightarrow\mathbb{E}[X_\infty(1-X_\infty)]=\frac{1}{4}-\frac{1}{36}=\frac{2}{9}.$$

Since $m_q\to1$, $m_q^2-4K_q\to1/9$. The Vandermonde part grows only like $-\frac{1}{2}\log q+O(1)$ by (49), while the displayed deficit grows linearly in q. ∎

What the cusp result does and does not prove

The family solves the purely geometric variational problem $\mathscr{S}<0$. It also proves that geometry alone is insufficient: a prime-support pattern still compatible with every inherited theorem makes A_N large enough to reverse the gain. A successful determinant family must couple the prime assignments intrinsically, rather than merely concentrate the real spectra.

8 A universal factorial divisor from integer-valued interpolation

The determinant method has so far used only prime factors already present in the entries. Integer evaluation matrices also acquire unavoidable "new-prime" divisors from finite-difference algebra. The next theorem is universal and appears not to have been used in the inherited report.

Definition 8.1. A finite set $\mathcal{S} \subset \mathbb{N}_0^2$ is a *coordinate lower set* if $(n,k) \in \mathcal{S}$ and $0 \le r \le n$, $0 \le s \le k$ imply $(r,s) \in \mathcal{S}$. Put

$$\mathfrak{F}(\mathcal{S}) = \prod_{(n,k)\in\mathcal{S}} n!\,k!. \tag{55}$$

Theorem 8.2 (Lower-set factorial divisor)**.** *Let $\mathcal{S} \subset \mathbb{N}_0^2$ be a lower set of cardinality N, and let $(X_i, Y_i) \in \mathbb{Z}^2$, $1 \le i \le N$. Then*

$$\mathfrak{F}(\mathcal{S}) \ \big|\ \det\big(X_i^n Y_i^k\big)_{1\le i\le N,\ (n,k)\in\mathcal{S}}. \tag{56}$$

Proof. Use the binomial basis $\binom{X}{r}\binom{Y}{s}$. The Stirling expansion is

$$X^n Y^k = \sum_{r\le n}\sum_{s\le k} \left\{ {n \atop r} \right\} \left\{ {k \atop s} \right\} r!s! \binom{X}{r}\binom{Y}{s}, \tag{57}$$

where the braces denote Stirling numbers of the second kind. Because $\mathcal{S}$ is lower, an order extending the coordinate partial order makes the column-change matrix triangular, with diagonal entries $n!k!$. The binomial evaluation matrix has integer entries for every integer X_i, Y_i. Taking determinants in (57) proves the divisibility. □

The theorem applies to the exact grid in two ways. If the row exponent set $\mathcal{R}$ is lower, transpose the evaluation matrix and use the integer points $(2^n M^k, 3^n A^k)$. If the column exponent set $\mathcal{C}$ is lower, use the integer points $(2^a 3^b, M^a A^b)$. Thus D_N is divisible by both $\mathfrak{F}(\mathcal{R})$ and $\mathfrak{F}(\mathcal{C})$, although their full product need not divide without additional hypotheses.

8.1 Exact growth on weighted triangles

Let $\mathcal{S}_N(w_1, w_2)$ be the first N lattice points in increasing order of $w_1 n + w_2 k$, with $w_1, w_2 > 0$. This is a lower set, up to an irrelevant choice among finitely many boundary ties.

Theorem 8.3 (Factorial asymptotic)**.** *As $N \to \infty$,*

$$\log \mathfrak{F}(\mathcal{S}_N(w_1,w_2)) = \frac{\sqrt{2}}{6}\frac{w_1 + w_2}{\sqrt{w_1 w_2}} N^{3/2} \log N + O_{w_1,w_2}(N^{3/2}). \tag{58}$$

If a fixed finite set of primes is deleted from the prime factorization of $\mathfrak{F}$, the leading term is unchanged.

Proof. The threshold R_N satisfies

$$N = \frac{R_N^2}{2w_1w_2} + O(R_N), \qquad R_N = \sqrt{2w_1w_2N} + O(1).$$

Stirling's formula and a lattice Riemann sum reduce the leading logarithmic term to one half of $\log N$ times the first coordinate moments:

$$\sum_{w_1n+w_2k\le R_N} (n+k) = \frac{R_N^3}{6}\left(\frac{1}{w_1^2w_2} + \frac{1}{w_1w_2^2}\right) + O(N)$$

$$= \frac{\sqrt{2}}{3} \frac{w_1 + w_2}{\sqrt{w_1 w_2}} N^{3/2} + O(N).$$

Multiplication by $\frac{1}{2} \log N$ gives (58); all remaining Stirling and boundary terms are $O(N^{3/2})$.

For a fixed prime p, Legendre's formula gives $v_p(n!) = n/(p-1) + O(\log n)$. Its total logarithmic contribution over the triangle is therefore $O_p(N^{3/2})$. Deleting finitely many primes changes only the error term. □

Let

$$\mathcal{P}_0 = \{p : p \mid 6MA\}.$$

All entry primes, and therefore all primes of Q_N, lie in $\mathcal{P}_0$. Write $\mathfrak{F}_{\mathcal{P}_0^c}$ for the part of $\mathfrak{F}$ supported outside $\mathcal{P}_0$.

Corollary 8.4 (Universal new-prime excess). *If $\mathcal{R}$ is lower, then*

$$\Xi_N = \log(D_N/Q_N) \geq \log \mathfrak{F}_{\mathcal{P}_0^c}(\mathcal{R}).$$

If both $\mathcal{R}$ and $\mathcal{C}$ are lower, the right side may be replaced by the larger of the corresponding row and column quantities. For the triangular grids,

$$\log \mathfrak{F}_{\mathcal{P}_0^c}(\mathcal{R}_N) = 0.483958920105405\ldots\ N^{3/2} \log N + O(N^{3/2}), \tag{59}$$

$$\log \mathfrak{F}_{\mathcal{P}_0^c}(\mathcal{C}_N) = \frac{\sqrt{2}}{6} \frac{1+\beta}{\sqrt{\beta}} N^{3/2} \log N + O_\beta(N^{3/2}). \tag{60}$$

At $\beta = 28$, the second coefficient is $1.291762669243385\ldots$.

Proof. Theorem 8.2 gives the divisibility of D_N. Outside $\mathcal{P}_0$ the tropical divisor Q_N has valuation zero, so the same prime powers divide D_N/Q_N. The constants follow from Theorem 8.3 with weights (α, γ) and $(1, \beta)$. □

A rigorous but subcritical gain

The factorial divisor is a genuine source of determinant primes not present in any matrix entry. Its size is nevertheless $N^{3/2} \log N = o(N^2)$. Since every critical geometric and HCIZ term is of order N^2, ordinary integer-valued interpolation cannot repair a positive leading screen. A successful arithmetic refinement must create N^2-scale divisibility or cancellation.

9 Exact signed-progression determinants and the denominator tax

Continued fractions suggest choosing exponent vectors along nearly level lines of the two logarithmic forms. Signed directions can make both row and column spacings tiny. The following family captures that idea exactly and, crucially, keeps every original exponent nonnegative.

Fix positive integers p, q, r, s and set, for $0 \leq i, j < N$,

$$(a_i, b_i) = (iq, (N-1-i)p), \qquad (n_j, k_j) = (jr, (N-1-j)s). \tag{61}$$

Put

$$X = 2^{qr}, \qquad Y = 3^{pr}, \qquad Z = M^{qs}, \qquad W = A^{ps}, \qquad P = XW, \quad Q = YZ. \tag{62}$$

Then

$$\log \frac{P}{Q} = (q\alpha - p\gamma)(r - s\beta). \tag{63}$$

Thus two independent rational approximations multiply their errors.

9.1 Complete factorization

Theorem 9.1 (Signed-progression factorization)**.** *For the matrix* (5) *built from* (61),

$$|D_N| = (PQ)^{\binom{N}{3}} \prod_{d=1}^{N-1} |P^d - Q^d|^{N-d}. \tag{64}$$

The determinant is nonzero because neither factor in (63) *vanishes.*

Proof. A direct expansion of the four exponents gives

$$E_{ij} = W^{(N-1)^2} \left(\frac{Z}{W}\right)^{(N-1)i} \left(\frac{Y}{W}\right)^{(N-1)j} \left(\frac{P}{Q}\right)^{ij}.$$

After extracting row and column factors, the remaining matrix is $(t^{ij})_{0\le i,j<N}$ with $t = P/Q$. Its Vandermonde determinant is

$$\det(t^{ij}) = t^{\binom{N}{3}} \prod_{d=1}^{N-1} (t^d - 1)^{N-d}. \tag{65}$$

Collecting the powers of Q yields exactly (64). The irrationalities $\alpha/\gamma \notin \mathbb{Q}$ and $\beta \notin \mathbb{Q}$ imply $q\alpha - p\gamma \neq 0$ and $r - s\beta \neq 0$. □

The monomial factor in (64) is a rational gauge artifact. Proposition 4.6 removes it completely. Let $g = \gcd(P, Q)$ and write

$$\widehat{P} = P/g, \qquad \widehat{Q} = Q/g, \qquad \gcd(\widehat{P}, \widehat{Q}) = 1.$$

Theorem 9.2 (Exact invariant local quotient)**.** *Let Q_N^{trop} be the termwise tropical divisor of the original integer matrix. Then*

$$\boxed{\frac{|D_N|}{Q_N^{\mathrm{trop}}} = \prod_{d=1}^{N-1} |\widehat{P}^d - \widehat{Q}^d|^{N-d}.} \tag{66}$$

In particular, every quantity on the right is an ordinary positive integer; all row, column, and common-gcd heights have already been paid.

Proof. By rational gauge invariance it suffices to calculate the quotient for (t^{ij}) with $t = \widehat{P}/\widehat{Q}$. If $v_\ell(t) > 0$, the reverse assignment minimizes $v_\ell(t)\sum_i i\sigma(i)$ and its value is $\binom{N}{3} v_\ell(\widehat{P})$. If $v_\ell(t) < 0$, the forward assignment maximizes the sum, giving $-(\sum_{i=0}^{N-1} i^2) v_\ell(\widehat{Q})$. Formula (65) shows that these are exactly the valuations of its pure numerator and denominator powers. Since $\widehat{P}^d - \widehat{Q}^d$ is coprime to both $\widehat{P}$ and $\widehat{Q}$, division by the local tropical factor leaves precisely the product in (66). □

9.2 Near-collision expansion

Set

$$\eta = \left|\log \frac{\widehat{P}}{\widehat{Q}}\right| > 0, \qquad B = \min(\widehat{P}, \widehat{Q}). \tag{67}$$

For $t \ge 0$, define $\rho(t) = \log((\mathrm{e}^t - 1)/t)$, with $\rho(0) = 0$. The elementary bounds $1 \le (\mathrm{e}^t - 1)/t \le \mathrm{e}^t$ give

$$0 \le \rho(t) \le t. \tag{68}$$

Theorem 9.3 (Exact denominator-tax expansion)**.** *The invariant quotient in* (66) *satisfies*

$$\log \frac{|D_N|}{Q_N^{\mathrm{trop}}} = \binom{N+1}{3} \log B + \binom{N}{2} \log \eta + \log G(N+1) + R_N, \tag{69}$$

where

$$0 \le R_N \le \binom{N+1}{3} \eta. \tag{70}$$

Moreover the integer gap gives

$$1 \le |\widehat{P} - \widehat{Q}| = B(\mathrm{e}^{\eta} - 1) \le B\eta \mathrm{e}^{\eta}, \qquad \log B + \log \eta \ge -\eta. \tag{71}$$

Consequently, for $0 < \eta \le 1$*,*

$$\log \frac{|D_N|}{Q_N^{\mathrm{trop}}} \ge \binom{N}{3} \log \frac{1}{\eta} - \binom{N+1}{3} \eta + \log G(N+1). \tag{72}$$

Proof. For each d,

$$|\widehat{P}^d - \widehat{Q}^d| = B^d(\mathrm{e}^{d\eta} - 1) = B^d d\eta\, \mathrm{e}^{\rho(d\eta)}.$$

Substitute this identity in (66). The three sums are

$$\sum_{d=1}^{N-1} d(N-d) = \binom{N+1}{3}, \quad \sum_{d=1}^{N-1} (N-d) = \binom{N}{2},$$

and $\sum_{d=1}^{N-1} (N-d) \log d = \log G(N+1)$. Bound the remainder with (68). The gap (71) follows because two distinct integers differ by at least one. Finally $\log B \ge -\log \eta - \eta$ in (69), and $\binom{N+1}{3} - \binom{N}{2} = \binom{N}{3}$. □

Corollary 9.4 (Why double continued fractions do not close the proof)**.** *Choose* p/q *among convergents to* α/γ *and* r/s *among convergents to* β*. Then*

$$\eta = |(q\alpha - p\gamma)(r - s\beta)| = O((qs)^{-1}).$$

Even after every rational gauge and common factor is removed, (72) *forces a positive cubic-in-*N *denominator cost of size at least*

$$\binom{N}{3} \log(qs) - O(N^3/(qs)) + \log G(N+1).$$

Thus the quadratic gain $\binom{N}{2} \log \eta$ *from the apparently tiny spacing is overpaid by the integer denominator needed to realize that spacing.*

The factorization also exposes all non-tropical valuations:

$$v_\ell(D_N) = \binom{N}{3} (v_\ell(P) + v_\ell(Q)) + \sum_{d=1}^{N-1} (N-d) v_\ell(P^d - Q^d). \tag{73}$$

For primes satisfying the standard hypotheses, the lifting-the-exponent lemma evaluates the last term explicitly. Hence any future use of this family must exploit structured cancellation inside cyclotomic factors $\widehat{P}^d - \widehat{Q}^d$; there is no unaccounted archimedean smallness left.

10 The fewnomial bridge to the Matrix Coefficient Conjecture

The rank-one logarithm matrix (3) is exactly the 2×2 case of the Matrix Coefficient Conjecture. Dasgupta and Kakde prove that their support condition (m') implies the desired rational coefficient vanishing, but the implication from singularity (w) to (m') remains open [3]. The integer-output hypothesis gives a particularly clean version of their zero-count mechanism.

Consider the transposed integer orbit

$$\mathcal{X}_L = \{(2^m 3^n, M^m A^n) : 0 \le m, n < L\} \subset \mathbb{N}^2. \tag{74}$$

Every point lies on the transcendental power curve $Y = X^\beta$, and all L^2 points are distinct.

Theorem 10.1 (Exact support lower bound)**.** *Assume a counterexample β exists. Let*

$$P(X,Y) = \sum_{(a,b)\in\mathcal{S}} c_{a,b} X^a Y^b \in \mathbb{R}[X,Y]$$

be nonzero, with every displayed coefficient nonzero. If P vanishes at all points of $\mathcal{X}_L$, then

$$|\mathcal{S}| \ge L^2 + 1. \tag{75}$$

In particular, vanishing on $\mathcal{X}_{2N}$ requires at least $4N^2 + 1$ monomials.

Proof. On the curve $Y = X^\beta$, put $X = \mathrm{e}^z$. The restriction becomes

$$P(\mathrm{e}^z, \mathrm{e}^{\beta z}) = \sum_{(a,b)\in\mathcal{S}} c_{a,b} \mathrm{e}^{(a+b\beta)z}. \tag{76}$$

The exponents $a + b\beta$ are distinct because β is irrational, so the right side is a nonzero exponential polynomial with $K = |\mathcal{S}|$ terms. Exponentials with distinct real exponents form a strict Chebyshev system and a nonzero K-term combination has at most $K - 1$ distinct real zeros. The values $z = m\alpha + n\gamma$, $0 \le m, n < L$, are distinct because $\alpha/\gamma \notin \mathbb{Q}$. Hence $K - 1 \ge L^2$. ☐

Corollary 10.2 (A concrete integer auxiliary-polynomial criterion)**.** *The Alaoglu–Erdős conjecture would follow if, under the assumption of a counterexample and for some N, one could construct a nonzero polynomial $P \in \mathbb{Z}[X,Y]$ with at most $4N^2$ monomials such that*

$$|P(2^m 3^n, M^m A^n)| < 1 \qquad (0 \le m, n < 2N). \tag{77}$$

The stronger Dasgupta–Kakde support target $|\operatorname{supp} P| < N^2$ is more than sufficient.

Proof. Every value in (77) is an integer, hence is zero. This contradicts Theorem 10.1. ☐

This criterion isolates the analytic heart of the problem. The zero estimate is elementary and sharp enough; the missing step is to manufacture a sparse integer polynomial whose archimedean values are all below one. Naive products that vanish for algebraic reasons have coefficient heights too large, while Siegel-lemma constructions with small support do not automatically produce uniform small values on the exponentially large grid. This is the same coefficient-height obstruction emphasized in the modern Matrix Coefficient program.

A potentially useful reformulation

Instead of asking first for exact vanishing, seek a support set $\mathcal{S}_N$ with $|\mathcal{S}_N| < N^2$ and integer coefficients c_w for which the normalized exponential sum

$$F_N(z) = \sum_{w=(a,b)\in\mathcal{S}_N} c_w \mathrm{e}^{(a+b\beta)z}$$

has a product-formula estimate

$$\log|F_N(m\alpha+n\gamma)|_\infty + \sum_p \log|F_N(m\alpha+n\gamma)|_p < 0$$

after clearing a common denominator, uniformly on $0 \le m, n < 2N$. Integrality would then force every value to vanish. The determinant factorizations of Sections 8 and 9 provide exact local factors that could be built into such a normalization.

11 Audit of additional proof routes

The supplied report explored many classical and computational directions. The new calculations above allow several of them to be classified more sharply.

Route	What it controls	Precise obstruction in the two-base problem
Baker–Matveev linear forms	Quantitative lower bounds for a *nonzero* linear form in logarithms of algebraic numbers	A counterexample gives the exact bilinear zero $\log M \log 3 - \log A \log 2 = 0$. Turning it into a fixed nonzero linear form loses the rank-two information.
Four Exponentials / Matrix Coefficient	Exactly the rational rank of the logarithm matrix	This would settle the conjecture, but the relevant 2×2 statement is itself the open Four Exponentials case. The sparse auxiliary-polynomial implication $(w) \Rightarrow (m')$ remains unknown.
Pure HCIZ asymptotics	The archimedean determinant free energy	For triangular grids, even the Jensen floor plus perfect joint tropical transport is positive beyond $\beta = 6.9269\ldots$; exact HCIZ evaluation alone cannot help.
Componentwise p-adic transport	Prime divisors inherited from individual entry factors	Primewise minimizing permutations can be incompatible. The resulting alignment defect A_N is nonnegative and is linear in the cusp parameter in an admissible separated-support branch.
Continued fractions	Very small projected row and column spacings	The invariant quotient formula (66) shows that integer denominators overpay the small spacing at cubic order.
Congruence collisions / finite differences	New divisibility from repeated residue classes and integer-valued bases	The universal factorial gain is only $N^{3/2}\log N$, below the critical N^2 scale. Fixed-modulus refinements have the same limitation.
Confluent determinants and jets	Replace close nodes by high multiplicity at one node	Derivatives of e^{uv} introduce powers of u and v, hence logarithms; the entries cease to be integers and the decisive product-formula gain is lost unless denominators are controlled adelically.

continued on next page

Audit of proof routes (continued)

Route	What it controls	Precise obstruction in the two-base problem
A third independent base	Six Exponentials supplies one transcendental value among a 2×3 array	If $2^x, 3^x, 5^x$ were all algebraic and $x \notin \mathbb{Q}$, Six Exponentials would contradict the $\mathbb{Q}$-independence of $\log 2, \log 3, \log 5$. The difficulty is intrinsically the critical two-base case.

11.1 Why a fixed finite computation cannot be the final proof

The finite verifier in Section 12 excludes a very large compact range, but a hypothetical primitive exponent has no known unconditional upper bound. The inherited monoid theorem creates infinitely many solutions from one primitive β, yet their integer first outputs grow as $2^n M^k$ and do not force a second primitive generator in a bounded interval. Consequently computation is valuable for raising the floor and testing constants, but a proof still needs a scale-free argument.

11.2 Why the new no-go theorems are useful

Negative results at the level of methods are not merely postmortems. They remove several tempting but insufficient tasks:

1. no amount of precision in the triangular HCIZ limit can bridge the joint screen gap;
2. no pair of continued fractions can make the exact progression quotient small without paying its denominator;
3. no ordinary binomial-basis divisibility argument can generate a leading N^2 term;
4. no cusp geometry can be universal unless it also enforces coupling of the four prime assignments.

The remaining work is therefore much more focused than at the start of the continuation.

12 Directed-rounding finite exclusion

This section extends the inherited finite computation while preserving a strict trust boundary. The mathematical reduction and the interpretation of a successful interval test are paper proofs. Exhausting hundreds of millions of inputs is a software certificate: it depends on the C compiler, MPFR, the runtime, and the recorded logs, and has not been replayed by the Lean kernel.

12.1 Reduction to consecutive-integer separation

Put

$$\vartheta = \frac{\log 3}{\log 2}. \tag{78}$$

If $m = 2^x \in \mathbb{N}$, then

$$3^x = m^\vartheta. \tag{79}$$

The input m is a power of two exactly when x is an integer. Therefore a nonintegral counterexample with $2^L \le m < 2^{L+1}$ would be a non-power of two for which m^ϑ is an integer.

Lemma 12.1 (Directed logarithmic certificate)**.** *Let $m, n \in \mathbb{N}$. If directed-rounding interval arithmetic proves*

$$\log m \, \log 3 > \log n \, \log 2, \qquad \log(n+1) \, \log 2 > \log m \, \log 3, \tag{80}$$

then

$$n < m^{\vartheta} < n+1,$$

so $m^{\vartheta} \notin \mathbb{Z}$.

Proof. All logarithms are positive. Divide (80) by $\log 2$ and exponentiate. □

The verifier first obtains a long-double locator $\lfloor \exp(\vartheta \log m) \rfloor$, but correctness does not depend on that approximation. It tests nearby candidates and counts an input as certified only when the two MPFR inequalities (80) are both strict. Thus an inaccurate locator could cause a reported *failure*, never a false success.

12.2 Implementation and reproducibility manifest

The literal verifier is reproduced in Appendix D. Its SHA-256 digest is

`f888c0d89be9366f01e2d26f339dcbe446556d39d4c5dbcf8f579ec0d0052e1d.`

The run environment was Linux x86-64, GCC 14.2.0, MPFR 4.2.2 linked against GMP, and five OpenMP threads on an Intel Xeon Platinum 8573C. Every transcendental operation and every product entering the certificate used outward MPFR rounding at 192 bits. The range was split into chunks of width 2^{22}; Appendix C lists every chunk.

The new range $[2^{27}, 2^{28})$ was scanned twice from the exact source. Both full scans had zero failures and identical checked counts, extremal margins, and extremal pairs; their logs differ only in timing data. The log digests are

`d8f782795864b6d6c4ec0a33fdbdbbb57ea46a8dd92fc20690f1ea5ab8ef1519,`
`a234067d7718c4ec638bb5d4a64ca3178159d118d69d70904654b6bd9749d4df.`

12.3 Completed dyadic ranges

range	chunks	nonpowers checked	failures	precision	provenance
$1 \le m < 2^{27}$	—	—	0	MPFR	inherited scans
$2^{27} \le m < 2^{28}$	32	134,217,727	0	192 bits	two full runs
$2^{28} \le m < 2^{29}$	64	268,435,455	0	192 bits	full run

The canonical $[2^{28}, 2^{29})$ log has SHA-256 digest

`b8822a77f9a46ad69a8d52849a80aca3526a63de3f76816e3cad6275d9cf2427.`

For $[2^{27}, 2^{28})$, the summed CPU-wall measurements printed by the 32 chunks were 716.927669 seconds. The smallest directed lower margin was

$$1.9529773015500319 \times 10^{-22} \quad \text{at} \quad (m, n) = (233{,}957{,}087, 18{,}397{,}848{,}948{,}879), \tag{81}$$

and the smallest directed upper margin was

$$5.4449255726026935 \times 10^{-22} \quad \text{at} \quad (m, n) = (183{,}569{,}533, 12{,}526{,}047{,}934{,}207). \tag{82}$$

At high precision, the corresponding ordinary gaps are

$$m^{\vartheta} - n = 5.1836871594112221948640711 79\ldots \times 10^{-9}, \tag{83}$$
$$n + 1 - m^{\vartheta} = 9.8396705106005791152768816 49\ldots \times 10^{-9}. \tag{84}$$

Both extremal inputs were independently replayed at 320 and 512 bits, with the same nearest integer and strictly positive directed margins.

For $[2^{28}, 2^{29})$, the aggregate results were

$$\text{summed chunk time} = 1468.193778 \text{ seconds}, \tag{85}$$
$$\text{min lower margin} = 4.9141462461940518 \times 10^{-24}, \quad \text{at } (m, n) = (388{,}909{,}543, 41{,}170{,}668{,}399{,}174), \tag{86}$$
$$\text{min upper margin} = 5.2103670350428438 \times 10^{-23}, \quad \text{at } (m, n) = (501{,}373{,}708, 61{,}578{,}600{,}753{,}798). \tag{87}$$

High-precision reevaluation gives

$$m^{\vartheta} - n = 2.9188416434683008767103555 34\ldots \times 10^{-10}, \tag{88}$$
$$n + 1 - m^{\vartheta} = 4.6288453654600268035074950 27\ldots \times 10^{-9}. \tag{89}$$

The two extremal inputs were replayed at 320 and 512 bits with unchanged nearest integers and positive directed margins. The concatenated replay records have SHA-256 digest

`afc53d5d5c20ca7f946310ae3dc1f8176f6c5af31eb175c9ff32326e4813919d.`

Theorem 12.2 (Software-certified exclusion through 2^{29})**.** *At the stated software-trust level,*

$$1 \le m < 2^{29}, \quad m^{\vartheta} \in \mathbb{N} \quad \Longrightarrow \quad m \text{ is a power of two.} \tag{90}$$

Consequently every nonintegral x satisfying $2^x, 3^x \in \mathbb{Z}$ obeys

$$\boxed{x > 29.} \tag{91}$$

Proof. The inherited scans cover $m < 2^{27}$, and the two completed new dyadic scans cover $2^{27} \le m < 2^{29}$. Every non-power input has a strict consecutive- integer certificate from Lemma 12.1. If $m = 2^x < 2^{29}$ came from a nonintegral solution, then (79) would make $m^{\vartheta} = 3^x$ an integer at a non-power input, a contradiction. The endpoint $m = 2^{29}$ corresponds to the integral exponent 29, so a nonintegral solution must satisfy $m > 2^{29}$ and $x > 29$. □

Trust boundary

Theorem 12.2 is not a theorem of Lean or of MPFR as a formal logic. It is a reproducible exhaustive computation whose source, hashes, chunk records, precision, and extreme cases are disclosed. The inherited kernel-checked floor remains $\beta \ge 12$ until a formal interval checker replays the certificates.

13 Three precise routes that would finish the conjecture

The continuation narrows the remaining problem to three mathematically explicit targets. They are not vague invitations to "improve the bounds": each is a sufficient theorem with a quantified threshold.

13.1 Route A: a negative screen with controlled defects

Proposition 13.1 (Determinant completion criterion)**.** *Assume a counterexample and suppose there is a sequence of exact-grid matrices for which, for some* $\varepsilon > 0$*,*

$$S_N \le -\varepsilon N^2, \qquad H_N + A_N = o(N^2). \tag{92}$$

Then the Alaoglu–Erdős conjecture follows. More generally it is enough that $\limsup_N (H_N + A_N)/N^2 < \varepsilon$*.*

Proof. The exact identity $\Xi_N = S_N + H_N + A_N$ would give $\Xi_N < 0$ for all sufficiently large N, contradicting the positive integrality of D_N/Q_N. □

The power cusps provide $S_N/N^2 < 0$ with arbitrarily large magnitude, but Theorem 7.4 shows that A_N need not be small. The next design problem is therefore not another radial concentration. It is to build exponent sets for which every prime cost is forced to use nearly the same reverse matching. Promising mechanisms include:

1. arithmetic coupling of row coordinates, so that a and b are not independently reorderable;
2. a multiscale geometry in which any permutation improving one prime incurs a larger loss at another prime;
3. an averaging over several determinant blocks whose alignment defects cancel or telescope while their negative screens add.

A theorem of the form $A_N \le C\, H_N + o(N^2)$ with a uniform C would already make the numerical cusp data actionable for sufficiently large q.

13.2 Route B: an analytic fewnomial

By Corollary 10.2, it is enough to derive from the rank-one logarithm relation a sparse integer polynomial with uniformly small values on $\mathcal{X}_{2N}$. A concrete target is:

Problem 13.2 (Integer $(w) \Rightarrow (m')$ target)**.** For every sufficiently large N, construct under the counterexample hypothesis a nonzero $P_N \in \mathbb{Z}[X, Y]$ such that

$$|\operatorname{supp} P_N| < N^2, \qquad \max_{0 \le m,n < 2N} |P_N(2^m 3^n, M^m A^n)| < 1. \tag{93}$$

The second inequality forces exact vanishing, and Theorem 10.1 then gives an immediate contradiction. The advantage over the general Matrix Coefficient setting is that the evaluation points and coefficients can both be integral. The disadvantage is that integrality leaves no room for a merely good approximation: the product formula must beat the coefficient height at every point simultaneously.

A potentially creative hybrid is to use a determinant cofactor vector as the coefficient vector of P_N. Sections 8 and 9 then provide exact common divisors of the cofactors. If those divisors reduce the primitive coefficient height by an $\exp(cN^2)$ factor, the remaining archimedean interpolation error could cross below one. The universal factorial divisor alone is too small, but a cyclotomic or many-prime refinement might reach the critical scale.

13.3 Route C: non-tropical valuation at quadratic scale

Termwise tropical divisibility uses only the smallest valuation among Leibniz terms. Determinants can have much larger valuation because many minimum terms cancel modulo powers of a prime. Let R_N be any explicitly proved divisor of D_N.

Proposition 13.3 (Quadratic divisor criterion)**.** *If, for some exact-grid family,*

$$\log R_N > V_N + C_N^+ + o(N^2), \tag{94}$$

then the conjecture follows. The same conclusion holds with C_N^+ replaced by any rigorous upper bound for $\log I_N$*.*

Proof. The HCIZ identity and $\log I_N \leq C_N^+$ give $\log D_N \leq V_N + C_N^+$. A positive integer cannot be divisible by an integer larger than itself. □

Theorem 8.3 supplies only $\log R_N \asymp N^{3/2} \log N$. To reach (94), one needs cancellation repeated across a positive proportion of the tropical layers. Formula (73) suggests studying the cyclotomic decomposition

$$\widehat{P}^d - \widehat{Q}^d = \prod_{e|d} \Phi_e(\widehat{P}, \widehat{Q})$$

together with weighted multiplicities $N - d$. A theorem forcing many small orders modulo many primes could create the missing N^2 contribution, but it must be uniform in the unknown integers M, A.

13.4 Formalization roadmap

The paper-level results most suitable for kernel checking are:

1. the algebraic factorization Theorems 9.1 and 9.2;
2. the binomial-basis divisor Theorem 8.2;
3. the support lower bound Theorem 10.1;
4. a small reflective checker for each MPFR certificate, replacing trust in the exhaustive C loop by a list of verified rational intervals.

The continuum asymptotics and HCIZ analysis are less urgent to formalize until an arithmetic alignment theorem makes them part of a closing argument.

14 Conclusion

The Alaoglu–Erdős conjecture remains unproved, but the continuation changes the shape of the search in several substantive ways.

First, every exact-grid determinant now has an exact accounting identity:

$$\log(D_N/Q_N) = S_N + H_N + A_N.$$

This prevents archimedean geometry, random-matrix entropy, and incompatible prime assignments from being conflated. Second, the inherited triangular family is decisively closed as a route: its best possible joint screen is positive far below the range already excluded. Third, power cusps prove that negative screens are realizable, while the separated-support theorem exposes the new arithmetic obstacle they create. Fourth, the exact progression quotient proves that double continued fractions do not supply free smallness; the integer denominator is visible term by term. Fifth, the lower-set factorial theorem identifies a universal source of new determinant primes and shows quantitatively why it is still subcritical. Sixth, the integer orbit turns the Dasgupta–Kakde fewnomial strategy into the explicit small-value criterion (93). Finally, the directed-rounding scan raises the software-certified floor to $x > 29$.

The most promising synthesis is no longer "one more determinant estimate." It is a construction that couples real transport to prime transport: either a negative-screen family with forced adelic alignment, or a sparse auxiliary polynomial whose coefficients inherit an N^2-scale determinant divisor. Both formulations attack the precise rank-two obstruction rather than a surrogate. They also provide clear falsification tests for future ideas: compute the screen, expose the denominator gauge, and account for every prime assignment before investing in a large asymptotic calculation.

Outcome and trust boundary

The requested conjectural proof has not been completed. The strongest new unconditional conclusions are the exact determinant decompositions, no-go theorems, support and divisibility results proved above. The strongest new finite conclusion is the reproducible software certificate $x > 29$. The report deliberately leaves the final theorem labelled open rather than turning a numerical or conjectural step into a false proof.

A Constants and elementary integrals

For reference, the constants used repeatedly in the report are

$$
\begin{aligned}
\alpha &= \log 2 = 0.69314718055994530 94\ldots, \\
\gamma &= \log 3 = 1.0986122886681096914\ldots, \\
J &= \frac{\pi}{8} - \frac{1}{3} = 0.0593657483653908215\ldots, \\
\delta_{\text{old}} &= \frac{11}{6} - \frac{\pi}{2} = 0.2625370065384367141\ldots, \\
\delta_{\text{sep}} &= \frac{16}{9} - \frac{\pi}{2} = 0.2069814509828811585\ldots, \\
\delta_{\text{joint}} &= \frac{4}{9} - \frac{\pi}{8} = 0.0517453627457202896\ldots.
\end{aligned}
$$

The beta integrals are

$$
\begin{aligned}
\int_0^1 \sqrt{s(1-s)}\,\mathrm{d}s &= \mathrm{B}(3/2, 3/2) = \frac{\pi}{8}, \\
\int_0^1 (1 - \sqrt{1-s})(1 - \sqrt{s})\,\mathrm{d}s &= \frac{\pi}{8} - \frac{1}{3}, \\
\int_0^1 [s(1-s)]^{1/q}\,\mathrm{d}s &= \mathrm{B}(1 + 1/q, 1 + 1/q).
\end{aligned}
$$

The combinatorial sums used in the progression determinant are

$$
\sum_{i=0}^{N-1} i(N-1-i) = \binom{N}{3},
$$

$$\sum_{d=1}^{N-1} d(N-d) = \binom{N+1}{3},$$
$$\sum_{d=1}^{N-1} (N-d) = \binom{N}{2},$$
$$\prod_{d=1}^{N-1} d^{N-d} = G(N+1).$$

B High-precision HCIZ reconnaissance code

The following minimal Python/mpmath program evaluates the quantile formula (36). For large q and N, hundreds or thousands of decimal digits are required because the totally positive exponential matrix is extremely ill-conditioned.

```
import mpmath as mp


def hciz_log_over_n2(N: int, c, q: int = 2, dps: int = 200):
    """Return (log I_N)/N^2 and the centered entropy H_N/N^2.

    q=2 gives the triangular radial law; general q gives density q*t^(q-1).
    The midpoint quantiles are used.
    """
    mp.mp.dps = dps
    c = mp.mpf(c)
    x = [((mp.mpf(i) + mp.mpf("0.5")) / N) ** (mp.mpf(1) / q)
         for i in range(N)]
    t = N * c
    A = mp.matrix([[mp.exp(t * x[i] * x[j]) for j in range(N)]
                   for i in range(N)])
    det_A = mp.det(A)
    if det_A <= 0:
        raise ArithmeticError("increase dps: determinant was not resolved")

    log_G = sum(mp.log(mp.factorial(j)) for j in range(1, N))
    log_delta = sum(mp.log(x[j] - x[i])
                    for i in range(N) for j in range(i + 1, N))
    pairs = N * (N - 1) // 2
    log_I = log_G - pairs * mp.log(t) + mp.log(det_A) - 2 * log_delta
    mean = sum(x) / N
    entropy = log_I / N**2 - c * mean**2
    return log_I / N**2, entropy


if __name__ == "__main__":
    alpha, gamma = mp.log(2), mp.log(3)
    beta = mp.mpf(28)
    z = mp.sqrt(beta * alpha * gamma)
    for q in (9, 10, 12, 16, 24):
        value, entropy = hciz_log_over_n2(32, q * z, q=q, dps=1600)
        print(q, mp.nstr(value, 20), mp.nstr(entropy, 20))
```

C Full directed-rounding chunk manifest

The following table records every newly executed chunk. The endpoints are half-open. “Lower” is the smallest certified value of $\log m \log 3 - \log n \log 2$ in the chunk; “upper” is the smallest certified value of $\log(n+1)\log 2 - \log m \log 3$.

Table 1: Canonical manifest of all newly executed directed-rounding chunks. Every row used MPFR 4.2.2 at 192 bits with five OpenMP threads.

chunk	half-open m-range	checked	failures	minimum lower margin	minimum upper margin	seconds
Dyadic block $2^{27} \leq m < 2^{28}$: 32 chunks, 134,217,727 nonpowers, zero failures.						
1	$[134, 217, 728, 138, 412, 032)$	$4, 194, 303$	0	4.2836769792074091e-20	3.3891031242020454e-20	21.573743
2	$[138, 412, 032, 142, 606, 336)$	$4, 194, 304$	0	2.814750234252554e-20	2.8468076218901581e-21	22.158163
3	$[142, 606, 336, 146, 800, 640)$	$4, 194, 304$	0	1.0607294660531761e-20	6.2725924488237467e-22	22.023830
4	$[146, 800, 640, 150, 994, 944)$	$4, 194, 304$	0	3.5925165762872837e-21	1.1433227292901367e-20	22.206484
5	$[150, 994, 944, 155, 189, 248)$	$4, 194, 304$	0	3.7822884041073772e-21	4.8277308838469829e-20	22.095471
6	$[155, 189, 248, 159, 383, 552)$	$4, 194, 304$	0	1.6774887404209645e-20	1.6063405167449774e-20	21.479050
7	$[159, 383, 552, 163, 577, 856)$	$4, 194, 304$	0	3.4689830423256351e-20	2.0205101475081749e-20	21.588538
8	$[163, 577, 856, 167, 772, 160)$	$4, 194, 304$	0	3.846788002095238e-21	7.5692748217377577e-21	22.685610
9	$[167, 772, 160, 171, 966, 464)$	$4, 194, 304$	0	8.6652029533225718e-21	1.4648482782275675e-20	22.715049
10	$[171, 966, 464, 176, 160, 768)$	$4, 194, 304$	0	4.2617695398058924e-20	1.4330594036020375e-20	22.047058
11	$[176, 160, 768, 180, 355, 072)$	$4, 194, 304$	0	1.8761737916482469e-20	4.4769685360901922e-21	22.485760
12	$[180, 355, 072, 184, 549, 376)$	$4, 194, 304$	0	5.6794633867480288e-20	5.4449255726026935e-22	23.087973
13	$[184, 549, 376, 188, 743, 680)$	$4, 194, 304$	0	3.085108520784044e-22	7.8715303310747852e-21	21.219970
14	$[188, 743, 680, 192, 937, 984)$	$4, 194, 304$	0	1.101691452511927e-20	1.7982910747464668e-20	21.240631
15	$[192, 937, 984, 197, 132, 288)$	$4, 194, 304$	0	1.6384741056980724e-21	2.1732258914807476e-20	23.096198
16	$[197, 132, 288, 201, 326, 592)$	$4, 194, 304$	0	8.7631120307607742e-21	1.1766250799413293e-20	22.451629
17	$[201, 326, 592, 205, 520, 896)$	$4, 194, 304$	0	9.3056083393033302e-22	9.2190395910340208e-21	21.764466
18	$[205, 520, 896, 209, 715, 200)$	$4, 194, 304$	0	2.7453613462399152e-21	1.3699056218129822e-21	23.099384
19	$[209, 715, 200, 213, 909, 504)$	$4, 194, 304$	0	1.3832865890950902e-20	8.3662430813457724e-21	23.155511
20	$[213, 909, 504, 218, 103, 808)$	$4, 194, 304$	0	4.10094165850518e-22	1.6787722485312616e-21	22.361308
21	$[218, 103, 808, 222, 298, 112)$	$4, 194, 304$	0	3.6380327784567066e-21	2.8732513111535305e-21	21.721567
22	$[222, 298, 112, 226, 492, 416)$	$4, 194, 304$	0	1.8253982235735446e-20	5.0528694537551156e-21	21.774815
23	$[226, 492, 416, 230, 686, 720)$	$4, 194, 304$	0	1.6457197628062163e-20	2.2280788319766659e-20	22.953286
24	$[230, 686, 720, 234, 881, 024)$	$4, 194, 304$	0	1.9529773015500319e-22	6.3472956707862069e-21	21.940421
25	$[234, 881, 024, 239, 075, 328)$	$4, 194, 304$	0	3.8124247870879489e-21	3.7502498645650887e-21	22.154232
26	$[239, 075, 328, 243, 269, 632)$	$4, 194, 304$	0	6.0463198212846063e-21	9.7239545340119667e-21	23.672616
27	$[243, 269, 632, 247, 463, 936)$	$4, 194, 304$	0	3.9709735440456608e-21	5.0056141181347373e-21	21.601095
28	$[247, 463, 936, 251, 658, 240)$	$4, 194, 304$	0	1.0774278287572064e-20	8.0518642488392324e-21	21.148179
29	$[251, 658, 240, 255, 852, 544)$	$4, 194, 304$	0	1.9287644098066004e-20	4.4205797737780912e-21	30.442443
30	$[255, 852, 544, 260, 046, 848)$	$4, 194, 304$	0	6.9900965838018964e-21	8.3073231820333117e-22	21.643209
31	$[260, 046, 848, 264, 241, 152)$	$4, 194, 304$	0	4.05129907666862e-21	9.4093233028399882e-22	21.609438
32	$[264, 241, 152, 268, 435, 456)$	$4, 194, 304$	0	1.3739807600268022e-21	4.1501174316744359e-21	21.730542
Dyadic block $2^{28} \leq m < 2^{29}$: 64 chunks, 268,435,455 nonpowers, zero failures.						
1	$[268, 435, 456, 272, 629, 760)$	$4, 194, 303$	0	1.4382942573212187e-20	1.2756766670555827e-20	21.959970
2	$[272, 629, 760, 276, 824, 064)$	$4, 194, 304$	0	1.2941797674316326e-20	1.3285211513910945e-20	21.791004
3	$[276, 824, 064, 281, 018, 368)$	$4, 194, 304$	0	1.3356602772611869e-21	1.4730693735280784e-20	22.244634
4	$[281, 018, 368, 285, 212, 672)$	$4, 194, 304$	0	6.2196805600968312e-21	2.8468076218901581e-21	23.106297
5	$[285, 212, 672, 289, 406, 976)$	$4, 194, 304$	0	3.5733784723795044e-21	1.5210349101083421e-20	21.901902
6	$[289, 406, 976, 293, 601, 280)$	$4, 194, 304$	0	8.5083827365367966e-21	6.2725924488237467e-22	27.162256
7	$[293, 601, 280, 297, 795, 584)$	$4, 194, 304$	0	3.5925165762872837e-21	3.2347592283203294e-21	21.489716
8	$[297, 795, 584, 301, 989, 888)$	$4, 194, 304$	0	1.7754344810408235e-21	3.2756881440037552e-20	21.688472
9	$[301, 989, 888, 306, 184, 192)$	$4, 194, 304$	0	1.7254677900998507e-21	3.2392687669939001e-21	21.687875
10	$[306, 184, 192, 310, 378, 496)$	$4, 194, 304$	0	5.190639911942908e-22	2.8791997145145888e-20	21.671935
11	$[310, 378, 496, 314, 572, 800)$	$4, 194, 304$	0	3.8871962533062486e-21	5.2038085056845994e-21	21.885574
12	$[314, 572, 800, 318, 767, 104)$	$4, 194, 304$	0	1.0482075541755741e-20	3.6531082881556569e-22	22.135351
13	$[318, 767, 104, 322, 961, 408)$	$4, 194, 304$	0	1.665998259948587e-20	2.0322875955387764e-21	23.413869
14	$[322, 961, 408, 327, 155, 712)$	$4, 194, 304$	0	1.3779479139462461e-20	2.0162681791702176e-20	22.309324
15	$[327, 155, 712, 331, 350, 016)$	$4, 194, 304$	0	9.2958235214673481e-21	1.9512358290838078e-21	21.783351
16	$[331, 350, 016, 335, 544, 320)$	$4, 194, 304$	0	3.846788002095238e-21	6.5115055825091894e-21	23.049868
17	$[335, 544, 320, 339, 738, 624)$	$4, 194, 304$	0	1.1129097627430943e-21	2.7432988422760654e-21	21.418060
18	$[339, 738, 624, 343, 932, 928)$	$4, 194, 304$	0	7.6501667219668186e-21	1.9891651868040581e-21	22.343562
19	$[343, 932, 928, 348, 127, 232)$	$4, 194, 304$	0	2.0000499711085253e-21	2.7716630995694901e-21	25.724149
20	$[348, 127, 232, 352, 321, 536)$	$4, 194, 304$	0	1.4372096305581895e-22	1.4330594036020375e-20	23.668225
21	$[352, 321, 536, 356, 515, 840)$	$4, 194, 304$	0	5.1428344200276079e-22	2.8329202469963484e-21	22.620369

continued on next page

Table 1 (continued)

chunk	half-open m-range	checked	failures	minimum lower margin	minimum upper margin	seconds
22	$[356,515,840,360,710,144)$	$4,194,304$	0	3.0805543496821567e-21	5.7354144573983494e-22	22.879868
23	$[360,710,144,364,904,448)$	$4,194,304$	0	8.1412921475699184e-22	1.3160134809143063e-22	21.562068
24	$[364,904,448,369,098,752)$	$4,194,304$	0	9.2401544639792503e-22	3.2227663898159427e-22	21.379615
25	$[369,098,752,373,293,056)$	$4,194,304$	0	7.1571294927829644e-22	8.104641440787022e-22	22.506453
26	$[373,293,056,377,487,360)$	$4,194,304$	0	3.085108520784044e-22	2.5290140084576452e-21	21.429379
27	$[377,487,360,381,681,664)$	$4,194,304$	0	8.8796853858061483e-21	6.4286500147541758e-21	21.480836
28	$[381,681,664,385,875,968)$	$4,194,304$	0	8.272911016338932e-22	1.0558727045286377e-20	22.288343
29	$[385,875,968,390,070,272)$	$4,194,304$	0	4.9141462461940518e-24	8.8567749855473731e-21	21.272564
30	$[390,070,272,394,264,576)$	$4,194,304$	0	1.6384741056980724e-21	8.1142647207276599e-22	21.301320
31	$[394,264,576,398,458,880)$	$4,194,304$	0	2.5262770232371374e-21	5.9115729230110735e-21	22.318805
32	$[398,458,880,402,653,184)$	$4,194,304$	0	7.884959025943028e-24	1.693989476820833e-20	22.435042
33	$[402,653,184,406,847,488)$	$4,194,304$	0	9.3056083393033302e-22	4.7107103532016623e-22	21.463475
34	$[406,847,488,411,041,792)$	$4,194,304$	0	9.6999134600105465e-22	2.8194555308677648e-21	21.617425
35	$[411,041,792,415,236,096)$	$4,194,304$	0	2.7453613462399152e-21	1.2931094234757654e-21	21.412518
36	$[415,236,096,419,430,400)$	$4,194,304$	0	5.2186368110621445e-21	7.6928727998249612e-22	21.715784
37	$[419,430,400,423,624,704)$	$4,194,304$	0	1.9540795171694777e-21	3.4356706045670469e-21	22.082776
38	$[423,624,704,427,819,008)$	$4,194,304$	0	1.2889645833520891e-21	2.5129329573956735e-22	21.482449
39	$[427,819,008,432,013,312)$	$4,194,304$	0	4.10094165850518e-22	4.882864438470783e-22	21.548963
40	$[432,013,312,436,207,616)$	$4,194,304$	0	5.9844399193589209e-21	1.6787722485312616e-21	43.612452
41	$[436,207,616,440,401,920)$	$4,194,304$	0	1.6849448054903945e-21	2.8732513111535305e-21	23.632709
42	$[440,401,920,444,596,224)$	$4,194,304$	0	1.2750297679026755e-21	5.0834429745619907e-21	22.749651
43	$[444,596,224,448,790,528)$	$4,194,304$	0	4.069770918056726e-21	7.1579665047950893e-21	23.374068
44	$[448,790,528,452,984,832)$	$4,194,304$	0	1.5687781985625409e-21	2.8217942789569643e-22	22.670732
45	$[452,984,832,457,179,136)$	$4,194,304$	0	4.4104212051744008e-21	1.0142750414892474e-21	22.095681
46	$[457,179,136,461,373,440)$	$4,194,304$	0	2.7933548125913395e-21	8.322373462350366e-21	23.014903
47	$[461,373,440,465,567,744)$	$4,194,304$	0	1.746726425117485e-21	6.3472956707862069e-21	22.372636
48	$[465,567,744,469,762,048)$	$4,194,304$	0	1.9529773015500319e-22	7.458651036202973e-22	21.847208
49	$[469,762,048,473,956,352)$	$4,194,304$	0	3.0673674405449177e-21	3.7502498645650887e-21	26.386328
50	$[473,956,352,478,150,656)$	$4,194,304$	0	3.2755905490807117e-21	2.1994289671237529e-22	22.374161
51	$[478,150,656,482,344,960)$	$4,194,304$	0	5.9343407514030539e-21	6.0858574582942566e-21	23.846169
52	$[482,344,960,486,539,264)$	$4,194,304$	0	4.3652926354364494e-22	4.6557688115456714e-22	23.785055
53	$[486,539,264,490,733,568)$	$4,194,304$	0	5.8454646178688984e-22	2.4686697308562258e-22	23.246242
54	$[490,733,568,494,927,872)$	$4,194,304$	0	6.9726149033058515e-21	1.4245923555867345e-21	25.107561
55	$[494,927,872,499,122,176)$	$4,194,304$	0	4.4549980557391829e-21	1.7924883453908774e-21	23.659180
56	$[499,122,176,503,316,480)$	$4,194,304$	0	5.7803937021546911e-22	5.2103670350428438e-23	22.775333
57	$[503,316,480,507,510,784)$	$4,194,304$	0	4.5327395047841313e-21	9.3695153236364644e-22	23.028648
58	$[507,510,784,511,705,088)$	$4,194,304$	0	1.1403749258280013e-21	1.1831074533612995e-21	23.076698
59	$[511,705,088,515,899,392)$	$4,194,304$	0	6.9900965838018964e-21	4.2736590198781848e-22	22.923524
60	$[515,899,392,520,093,696)$	$4,194,304$	0	1.1153565892567642e-21	8.3073231820333117e-22	24.388771
61	$[520,093,696,524,288,000)$	$4,194,304$	0	6.0229819895868463e-22	9.4093233028399882e-22	22.257219
62	$[524,288,000,528,482,304)$	$4,194,304$	0	1.3256677749805584e-21	1.9721625002617089e-21	22.349126
63	$[528,482,304,532,676,608)$	$4,194,304$	0	4.3601564458959503e-21	9.3067114720708575e-22	24.270259
64	$[532,676,608,536,870,912)$	$4,194,304$	0	1.3739807600268022e-21	4.7528597804668423e-21	22.116018

D Directed-rounding C source

This is the exact source whose digest is printed in Section 12.

```c
#define _GNU_SOURCE
#include <stdio.h>
#include <stdlib.h>
#include <stdint.h>
#include <inttypes.h>
#include <math.h>
#include <omp.h>

#if defined(__has_include)
# if __has_include(<mpfr.h>)
# include <mpfr.h>
# define HAVE_SYSTEM_MPFR_HEADER 1
# endif
#endif

#ifndef HAVE_SYSTEM_MPFR_HEADER
/* Minimal declarations for the MPFR 4.x ABI. The execution environment had
   libmpfr.so.6 but not the development header package. A normal build with
   libmpfr-dev installed takes the branch above instead. */
typedef unsigned long mp_limb_t;
typedef long mpfr_prec_t;
typedef long mpfr_exp_t;
typedef int mpfr_sign_t;
typedef struct {
  mpfr_prec_t _mpfr_prec;
  mpfr_sign_t _mpfr_sign;
  mpfr_exp_t _mpfr_exp;
  mp_limb_t *_mpfr_d;
} __mpfr_struct;
typedef __mpfr_struct mpfr_t[1];
typedef __mpfr_struct *mpfr_ptr;
typedef const __mpfr_struct *mpfr_srcptr;
typedef enum {
  MPFR_RNDN = 0, MPFR_RNDZ = 1, MPFR_RNDU = 2,
  MPFR_RNDD = 3, MPFR_RNDA = 4, MPFR_RNDF = 5,
  MPFR_RNDNA = -1
} mpfr_rnd_t;
extern void mpfr_init2(mpfr_ptr, mpfr_prec_t);
extern void mpfr_clear(mpfr_ptr);
extern int mpfr_set_ui(mpfr_ptr, unsigned long, mpfr_rnd_t);
extern int mpfr_log(mpfr_ptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_mul(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_sub(mpfr_ptr, mpfr_srcptr, mpfr_srcptr, mpfr_rnd_t);
extern int mpfr_cmp_ui(mpfr_srcptr, unsigned long);
extern double mpfr_get_d(mpfr_srcptr, mpfr_rnd_t);
extern const char *mpfr_get_version(void);
#endif

static inline int is_power_of_two_u64(uint64_t x) {
  return x && ((x & (x - 1)) == 0);
}

typedef struct {
  uint64_t checked;
  uint64_t failures;
  double min_lower;
  double min_upper;
  uint64_t min_lower_m, min_lower_n;
  uint64_t min_upper_m, min_upper_n;
  uint64_t first_fail_m, first_fail_guess;
} Stats;

int main(int argc, char **argv) {
  uint64_t start = 1;
  uint64_t limit = (1ULL << 20);
  int bits = 192;
  if (argc > 1) start = strtoull(argv[1], NULL, 0);
  if (argc > 2) limit = strtoull(argv[2], NULL, 0);
```

```
  if (argc > 3) bits = atoi(argv[3]);
  if (start < 1 || limit <= start) {
    fprintf(stderr, "bad range\n");
    return 2;
  }

  int nt = omp_get_max_threads();
  Stats *stats = calloc((size_t)nt, sizeof(Stats));
  if (!stats) return 3;
  for (int i = 0; i < nt; ++i) {
    stats[i].min_lower = INFINITY;
    stats[i].min_upper = INFINITY;
  }

  const long double theta = logl(3.0L) / logl(2.0L);
  double t0 = omp_get_wtime();

  mpfr_t two, three, ln2_lo, ln2_hi, ln3_lo, ln3_hi;
  mpfr_init2(two, bits); mpfr_init2(three, bits);
  mpfr_init2(ln2_lo, bits); mpfr_init2(ln2_hi, bits);
  mpfr_init2(ln3_lo, bits); mpfr_init2(ln3_hi, bits);
  mpfr_set_ui(two, 2, MPFR_RNDN);
  mpfr_set_ui(three, 3, MPFR_RNDN);
  mpfr_log(ln2_lo, two, MPFR_RNDD);
  mpfr_log(ln2_hi, two, MPFR_RNDU);
  mpfr_log(ln3_lo, three, MPFR_RNDD);
  mpfr_log(ln3_hi, three, MPFR_RNDU);

#pragma omp parallel
  {
    int tid = omp_get_thread_num();
    Stats *st = &stats[tid];
    mpfr_t xm, xn, xnp1;
    mpfr_t logm_lo, logm_hi, logn_hi, lognp1_lo;
    mpfr_t lhs_lo, lhs_hi, rhs_lo, rhs_hi, dl, du;
    mpfr_init2(xm, bits); mpfr_init2(xn, bits); mpfr_init2(xnp1, bits);
    mpfr_init2(logm_lo, bits); mpfr_init2(logm_hi, bits);
    mpfr_init2(logn_hi, bits); mpfr_init2(lognp1_lo, bits);
    mpfr_init2(lhs_lo, bits); mpfr_init2(lhs_hi, bits);
    mpfr_init2(rhs_lo, bits); mpfr_init2(rhs_hi, bits);
    mpfr_init2(dl, bits); mpfr_init2(du, bits);

#pragma omp for schedule(static)
    for (uint64_t m = start; m < limit; ++m) {
      if (is_power_of_two_u64(m)) continue;

      /* This is only a locator. The MPFR inequalities below are the proof. */
      uint64_t guess = (uint64_t)floorl(expl(theta * logl((long double)m)));

      mpfr_set_ui(xm, (unsigned long)m, MPFR_RNDN);
      mpfr_log(logm_lo, xm, MPFR_RNDD);
      mpfr_log(logm_hi, xm, MPFR_RNDU);
      mpfr_mul(lhs_lo, logm_lo, ln3_lo, MPFR_RNDD);
      mpfr_mul(lhs_hi, logm_hi, ln3_hi, MPFR_RNDU);

      int ok = 0;
      uint64_t certn = 0;
      double dl_d = 0.0, du_d = 0.0;
      static const int offsets[7] = {0, -1, 1, -2, 2, -3, 3};
      for (int oi = 0; oi < 7; ++oi) {
        int off = offsets[oi];
        if (off < 0 && guess < (uint64_t)(-off) + 1) continue;
        uint64_t n = (uint64_t)((int64_t)guess + off);
        if (n < 1) continue;

        mpfr_set_ui(xn, (unsigned long)n, MPFR_RNDN);
        mpfr_set_ui(xnp1, (unsigned long)(n + 1), MPFR_RNDN);

        /* Directed lower bound for log(m)log(3)-log(n)log(2). */
        mpfr_log(logn_hi, xn, MPFR_RNDU);
        mpfr_mul(rhs_hi, logn_hi, ln2_hi, MPFR_RNDU);
        mpfr_sub(dl, lhs_lo, rhs_hi, MPFR_RNDD);
        if (mpfr_cmp_ui(dl, 0) <= 0) continue;
```

```

      /* Directed lower bound for log(n+1)log(2)-log(m)log(3). */
      mpfr_log(lognp1_lo, xnp1, MPFR_RNDD);
      mpfr_mul(rhs_lo, lognp1_lo, ln2_lo, MPFR_RNDD);
      mpfr_sub(du, rhs_lo, lhs_hi, MPFR_RNDD);
      if (mpfr_cmp_ui(du, 0) <= 0) continue;

      ok = 1;
      certn = n;
      dl_d = mpfr_get_d(dl, MPFR_RNDD);
      du_d = mpfr_get_d(du, MPFR_RNDD);
      break;
    }

    if (!ok) {
      ++st->failures;
      if (!st->first_fail_m || m < st->first_fail_m) {
        st->first_fail_m = m;
        st->first_fail_guess = guess;
      }
      continue;
    }

    ++st->checked;
    if (dl_d < st->min_lower) {
      st->min_lower = dl_d;
      st->min_lower_m = m;
      st->min_lower_n = certn;
    }
    if (du_d < st->min_upper) {
      st->min_upper = du_d;
      st->min_upper_m = m;
      st->min_upper_n = certn;
    }
  }

  mpfr_clear(xm); mpfr_clear(xn); mpfr_clear(xnp1);
  mpfr_clear(logm_lo); mpfr_clear(logm_hi);
  mpfr_clear(logn_hi); mpfr_clear(lognp1_lo);
  mpfr_clear(lhs_lo); mpfr_clear(lhs_hi);
  mpfr_clear(rhs_lo); mpfr_clear(rhs_hi);
  mpfr_clear(dl); mpfr_clear(du);
}

Stats total = {0};
total.min_lower = INFINITY;
total.min_upper = INFINITY;
for (int i = 0; i < nt; ++i) {
  Stats *s = &stats[i];
  total.checked += s->checked;
  total.failures += s->failures;
  if (s->min_lower < total.min_lower) {
    total.min_lower = s->min_lower;
    total.min_lower_m = s->min_lower_m;
    total.min_lower_n = s->min_lower_n;
  }
  if (s->min_upper < total.min_upper) {
    total.min_upper = s->min_upper;
    total.min_upper_m = s->min_upper_m;
    total.min_upper_n = s->min_upper_n;
  }
  if (s->first_fail_m &&
      (!total.first_fail_m || s->first_fail_m < total.first_fail_m)) {
    total.first_fail_m = s->first_fail_m;
    total.first_fail_guess = s->first_fail_guess;
  }
}

double elapsed = omp_get_wtime() - t0;
printf("mpfr_version=%s\n", mpfr_get_version());
printf("range=[%" PRIu64 ",%" PRIu64 ")\n", start, limit);
printf("precision_bits=%d\nthreads=%d\n", bits, nt);
printf("checked_nonpowers=%" PRIu64 "\n", total.checked);
```

```
  printf("failures=%" PRIu64 "\n", total.failures);
  printf("min_lower_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
         total.min_lower, total.min_lower_m, total.min_lower_n);
  printf("min_upper_log_margin=%.17g at (m,n)=(%" PRIu64 ",%" PRIu64 ")\n",
         total.min_upper, total.min_upper_m, total.min_upper_n);
  if (total.failures) {
    printf("first_failure=(m,guess)=(%" PRIu64 ",%" PRIu64 ")\n",
           total.first_fail_m, total.first_fail_guess);
  }
  printf("elapsed_seconds=%.6f\n", elapsed);

  mpfr_clear(two); mpfr_clear(three);
  mpfr_clear(ln2_lo); mpfr_clear(ln2_hi);
  mpfr_clear(ln3_lo); mpfr_clear(ln3_hi);
  free(stats);
  return total.failures ? 1 : 0;
}
```

References

[1] L. Alaoglu and P. Erdős, On highly composite and similar numbers, *Transactions of the American Mathematical Society* **56** (1944), 448–469. DOI: 10.2307/1990319.

[2] A. Baker, Linear forms in the logarithms of algebraic numbers, *Mathematika* **13** (1966), 204–216.

[3] S. Dasgupta and M. Kakde, Ranks of matrices of logarithms of algebraic numbers II: the matrix coefficient conjecture, *Advances in Mathematics* **487** (2026), Article 110753. DOI: 10.1016/j.aim.2025.110753; arXiv:2408.08178.

[4] A. O. Gel'fond, *Transcendental and Algebraic Numbers*, Dover, New York, 1960.

[5] Harish-Chandra, Differential operators on a semisimple Lie algebra, *American Journal of Mathematics* **79** (1957), 87–120.

[6] C. Itzykson and J.-B. Zuber, The planar approximation. II, *Journal of Mathematical Physics* **21** (1980), 411–421.

[7] S. Karlin, *Total Positivity*, Vol. I, Stanford University Press, 1968.

[8] L. Fousse, G. Hanrot, V. Lefèvre, P. Pélissier, and P. Zimmermann, MPFR: a multiple-precision binary floating-point library with correct rounding, *ACM Transactions on Mathematical Software* **33** (2007), Article 13.

[9] M. Waldschmidt, The Four Exponentials Problem and Schanuel's Conjecture, in *Mathematics Going Forward*, Lecture Notes in Mathematics **2313**, Springer, 2023, pp. 579–592. DOI: 10.1007/978-3-031-12244-6_39.